PSC REF#:546537

Public Service Commission of Wisconsin
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ENVIRONMENTAL ASSESSMENT

of

Wisconsin Electric Gas Operations
Rochester Lateral Project

Docket 6630-CG-139

Application of Wisconsin Electric Power Company-Gas Operations d/b/a We Energies, a Gas Public Utility, for Authority to Install Natural Gas Transmission Facilities in the Towns of Brighton, Dover, and Norway, the Villages of Rochester, Raymond, and Caledonia, and the City of Oak Creek, in Kenosha, Racine, and Milwaukee Counties, Wisconsin

April 2025

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Public Service Commission of Wisconsin

Division of Digital Access, Consumer, and Environmental Affairs

Office of Environmental Analysis

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Acronyms

§	Section
AHI	Architecture History Inventory
AIS	Agricultural Impact Statement
AMP	Agricultural Mitigation Plan
APE	Area of Potential Effect
ASI	Archaeological Site Inventory
ASNRI	Areas of special natural resource interest
BMP	Best management practices
Commission	Public Service Commission of Wisconsin
DATCP	Department of Agriculture, Trade, and Consumer Protection
dB	Decibels
dbh	Diameter at breast height
DNR	Wisconsin Department of Natural Resources
EA	Environmental Assessment
EDF	Environmental Defense Fund
EIS	Environmental Impact Statement
ERGS	Elm Road Generating Station
ER Review	Certified Endangered Resources Review
ER	Endangered resources
ERF	Electronic Records Filing System
ESA	Endangered Species Act
FCL	Forest Crop Law
FEMA	Federal Emergency Management Agency
GHG	Greenhouse gas
HPO	Historic Preservation Officer
IAM	Independent Agricultural Monitor
IEM	Independent Environmental Monitor
IR	Inadvertent release
LNG	Liquefied natural gas
LWCF	Land and Water Conservation Fund
MAOP	Maximum Allowable Operating Pressure
MFL	Managed Forest Law
MISO	Midcontinent Independent System Operator
HDD	Horizontal directional drilling
NAIP	National Agriculture Imagery Program
NHI	Natural Heritage Inventory
NFHL	National Flood Hazard Layer
NRCS	Natural Resource Conservation Survey
NRHP	National Register of Historic Places
OCCT	Oak Creek Combustion Turbine
OCLNG	Oak Creek liquified natural gas

OHWM	Ordinary high-water mark
psig	Pounds per square inch gauge
PSC	Public Service Commission
PTD	Permission to disturb
RICE	Reciprocating Internal Combustion Engine
RLP	Rochester Lateral Project
ROW	Right-of-way
RPBB HPZ	Rusty Patched Bumble Bee High Potential Zone
SCCT	Simple-cycle combustion turbine
SHPO	State Historic Preservation Office
TCSB	Temporary clear span bridge
USEPA	U.S. Environmental Protection Agency
USFWS	U.S. Fish and Wildlife Service
WE-GO	Wisconsin Electric Gas Operations
WHO	World Health Organization
WHPD	Wisconsin Historic Preservation Database
WICCI	Wisconsin Initiative on Climate Change Impacts
Wis. Admin. Code	Wisconsin Administrative Code
WisDOT	Wisconsin Department of Transportation
WWI	Wisconsin Wetland Indicator

1. Introduction

Wisconsin Electric Gas Operations (WE-GO or applicant) filed an application with the Public Service Commission of Wisconsin (PSC or Commission) on October 1, 2024, seeking authority to install approximately 33 miles of 24-inch and 30-inch steel 650 pounds per square inch gauge (psig) maximum allowable operating pressure (MAOP) natural gas transmission main in the towns of Brighton, Dover, and Norway and the villages of Rochester, Raymond and Caledonia, and the city of Oak Creek, in Kenosha, Racine, and Milwaukee Counties. This project is called the "Rochester Lateral Project" or "RLP". The proposed project is described as having two potential routes, "Route A" and "Route B". The estimated total cost for Route A is $210,942,310 and for Route B is $215,671,562. A third route, Route C, is also mentioned in the application. Route C consists of segments that could be used for connecting Route A and Route B.

The Commission requires applicants to file applications according to Commission prescribed filing requirements. The requirements were created to ensure the Commission receives every piece of information it needs to assist in the comprehensive review and analysis of all the merits of a proposed project, including environmental review. The requirements are based on statutory and administrative code, along with experience gained through prior project review.

1.1 Analysis for Wisconsin Environmental Policy Act Compliance

Wisconsin Admin. Code § PSC 133.03(1)(e) requires that a public gas utility obtain Commission authorization under Wis. Stat. § 196.49(5g)(ar)(d) for any single gas pipeline project, the cost of which exceeds $5,000,000. This is a Type II action under Wis. Admin. Code § PSC 4.10(2). For a Type II action, the Commission must prepare an Environmental Assessment (EA) to consider and document the environmental impacts of the project. When the EA is complete, a preliminary determination is made on whether to undertake a full Environmental Impact Statement (EIS), and comments on that determination are considered before a final determination is made. At the time of the preliminary determination, the Commission makes copies of the EA available to persons that request it.

An EA is a written review of the potential impacts of the proposed project that would affect the quality of the human environment as described in Wis. Stat. § 1.11(2)(c). An EA also describes ways of mitigating or avoiding some of the expected impacts and concludes with the evaluation of ten items described in Wis. Admin. Code § PSC 4.20(2)(d).

In accordance with Wis. Admin. Code § PSC 4.20(1m), notification of the Commission's intent to prepare an EA, including a solicitation for comments on the environmental aspects of this proposed project, was sent to the mailing list for this docket on November 20, 2024. The mailing list includes:

- Local residents and landowners potentially affected by the project
- Municipal officials in the towns and counties covered by the project area
- Local news media

- Libraries in the project area
- Regional Planning Commissions
- Legislators representing the affected area
- Any other persons with a demonstrated interest in the proposed project

Through the EA scoping period, Commission staff solicit public comments about the proposed project and take any comments of concerns regarding the environmental assessment or review of the project into consideration during the analysis of the project.

1.2 Environmental Assessment Scope

Wisconsin Admin. Code § PSC 4.20(1) requires that an EA be a concise document that provides a factual investigation of the relevant areas of environmental concern in sufficient depth to permit a reasonably informed preliminary judgement of the environmental consequences of the proposed project. An EA must include a recommendation whether the proposed project is a major action significantly affecting the quality of the human environment, within the meaning of Wis. Stat § 1.11(2)(c), for which an EIS is required. An EIS is required if an EA determines there are significant impacts to the environment as a result of the project.

The scope of an EA includes the construction impacts that are likely to occur for the site under consideration. The scope of an EA also discusses impacts that could be expected during the operational life of the project.

The Commission's Division of Digital Access, Consumer, and Environmental Affairs prepared this EA in cooperation with the Wisconsin Department of Natural Resources (DNR) Office of Energy to determine if an EIS is necessary under Wis. Stat. § 1.11.

A preliminary determination was made on April 29, 2025, concluding that preparation of an EIS was not necessary.[1]

This EA is being submitted as an exhibit in the technical hearing for the docket.

1.3 Information Received During EA Process

Wisconsin Admin. Code § PSC 4.20(2)(f) requires that the EA include a list of other persons contacted and a summary of comments.

1.3.1 Contributors to this EA

No other persons besides staff at the DNR and Commission were contacted or involved in the preparation of this EA. The following DNR and Commission staff contributed to the EA:

- Macaulay Haller, DNR Environmental Analysis and Review Specialist, provided information about wetlands and waterways and wrote those EA sections.

[1] PSC REF#: 543711 EA – Preliminary Determination Letter

- Stacy Rowe, DNR Conservation Biologist, Bureau of Environmental Analysis and Sustainability, provided information about potential impacts to endangered resources and wrote that EA section.

- Emily Thompson, PSC Environmental Analysis and Review Specialist, prepared the EA.

- Tyler Tomaszewski, PSC Environmental Analysis and Review Specialist, assisted with Geographic Information Systems (GIS) analysis and maps.

- Andrew Craft, PSC Environmental Analysis and Review Specialist, assisted with historic and cultural resources EA section.

- Evan Uphoff, PSC Public Service Engineer, assisted with EA sections on technical information and project purpose and need.

1.3.2 Summary of Public Comments

EA Notification Comment Period

A total of 15 public comments were received during the EA notification period. All comments were reviewed and considered during the EA scoping period and in the preparation of this EA. All comments are publicly available and can be accessed via the Case Management System (CMS). A list of public comments received during the EA notification period can be found in Appendix A of this EA.

One respondent expressed general support for the project, stating that they would like to have a natural gas pipeline on their road.

The remaining 14 respondents opposed the project or expressed concerns about aspects of the project. Concerns regarding the project include, but are not limited to, proximity of the natural gas pipeline to residential areas; gas leaks or explosions; the number of trees that would need to be cleared; impacts to wetlands, endangered species, bodies of water, and other natural areas; groundwater contamination; removal of private fences and landscaping; impacts to property values; disturbances to horses; and impacts to climate change and human health. The most common concerns included gas leaks/explosions, damage to wildlife and ecosystems, and proximity to residences.

Several landowners submitted public comments expressing concern about the gas pipeline being too close to their house, and the impacts the construction would have on their property – including trees, fences, and landscaping. Each of these properties was researched to view the extent of disturbance on their land and data requests were sent to the applicant for confirmation. These properties included segment A-8 north of Foley Road, segment A-6 on 43rd Street, segment A-6 on Seven Mile Road, segment A-6 along 76th Street, segment B-6 north of Forest Hills Road, and segment B-7 along West River Road. One respondent included a petition with signatures of landowners along Sunnyside Drive that oppose Route A due to its proximity to

houses. The company proposed segments C8 and C9 in this area to reduce impacts to residences along Sunnyside Drive. Another respondent owns agricultural land that the pipeline would run through and was concerned about the company following best management practices (BMPs) and crop yield loss that would be experienced post-construction.

Another respondent suggested to locate the pipeline on land that We Energies already owns, and to additionally move pipeline segment A-6 to the north side of Gorney Park.

Phantom Fireworks submitted a comment expressing concern over construction restricting their business operations during April – July.

All comments were reviewed and considered. Commission staff inquired through Data Requests where feasible route modifications were suggested in order to avoid or mitigate impacts to private properties.

Preliminary Determination (PD) Letter Comment Period

A total of 19 public comments were received during the PD comment period. Main issues raised included concern about the pipeline crossing private property; wanting to phase out of natural gas and use renewable options instead; negative impacts this project would have to the environment and climate change; and concern about rate increases. Many comments requested that the PSC conduct an EIS for this project for these reasons and others.

Walnut Way Conservation Corp. expressed the need for an EIS to be created for this project. Another comment submitted included views from seven different health, justice, and environmental groups, including, but not limited to, Sierra Club Wisconsin, Healthy Climate Wisconsin, and Wisconsin Environmental Justice & Infrastructure Initiative. All comments were reviewed and considered.

Additional public comment periods are available during the public hearing for this project.

A list of public comments received during the PD letter comment period can be found in Appendix B of this EA.

2. Project Purpose and Need

WE-GO states in its application that the purpose of the proposed project is to provide additional firm deliverability of natural gas to southeastern Wisconsin which would, in part, provide additional required firm natural gas service to Wisconsin Electric's proposed Oak Creek Combustion Turbine (OCCT) generation facility (docket 6630-CE-317), the proposed Paris Reciprocating Internal Combustion Engine (RICE) generation facility (docket 6630-CE-316), and subsequently to the Elm Road Generating Station (ERGS) after enhancements are made that will allow ERGS to operate completely fueled by natural gas.

The applicant provided forecasted firm load requirements which exceed its current firm load requirements. The ERGS, OCCT, and Paris RICE projects represent nearly all of the increased natural gas demand on the applicant's system; other customer demand growth represents a small fraction. The applicant projected that loads on its gas system will exceed currently contracted supply sources for between 35 and 50 days each winter. In conjunction with additional pipeline capacity from ANR Pipeline Company's 2027 capacity expansion and the OCLNG project proposed in docket 6630-CG-140, the applicant designed the RLP project to meet estimated peak winter demand.

The level of demand from the gas-fired electric generation facilities assumes the electric facilities will need to meet Midcontinent Independent System Operator, Inc.'s (MISO) Resource Adequacy construct, including reliability during winter months. The estimated demand of the electric generation plants assumes operation at their winter capacity factors in a winter scenario where electric demand relies on dispatchable gas-fired generation, in lieu of reliable renewable electric generation.

WE-GO further states in its application:

> The additional generation that will be provided by the OCCT Project is needed to ensure reliability and resiliency in the face of evolving regional energy market resource adequacy rules established by the Midcontinent Independent System Operator (MISO), manage substantial load growth, and ensure compliance with proposed US Environmental Protection Agency (USEPA) rules.
>
> The Company continues to focus on meeting customer demand for natural gas service. Recently, demand for firm natural gas service has increased significantly in southeastern Wisconsin. The increased demand for firm natural gas service was analyzed and it was determined that an approach including an increase in capacity on the Company's local distribution network and local natural gas storage in the form of liquefied natural gas (LNG) provides the optimal solution in terms of economics and reliability.

3. Project Description

As noted above, WE-GO is proposing to install approximately 33 miles of 24-inch and 30-inch steel 650 psig MAOP natural gas transmission main. The 30-inch main would extend from the existing Rochester Gate Station to the Oak Creek Power Plant site. The 24-inch main would extend from the proposed 30-inch main to the Lakeshore Lateral. Approximately seven percent of the gas pipeline would be located within public road right-of-way (ROW) for Route A. Along Route B, approximately five percent of the pipeline would be located within public road ROW. Some private land would be utilized for the gas pipeline ROW.

Additionally, the proposed project would also include the following associated infrastructure:

- Installation of five valve assemblies
- Improvements to an existing gate station
- Installation of a new district regulator station
- Replacement of 2,287 feet of existing 16-inch main with 2,833 feet of 24-inch main
- Installation of a 126 foot 6-inch and a 126 foot 20-inch main to the proposed OCLNG facility
- Installation of a 255 foot 20-inch service and turbine meters to serve the proposed OCCT at the simple-cycle combustion turbine (SCCT) meter set.
- Installation of a 90 foot 2-inch service and rotary meter to serve the water bath heaters and generator to serve the proposed OCCT at the SCCT meter set.

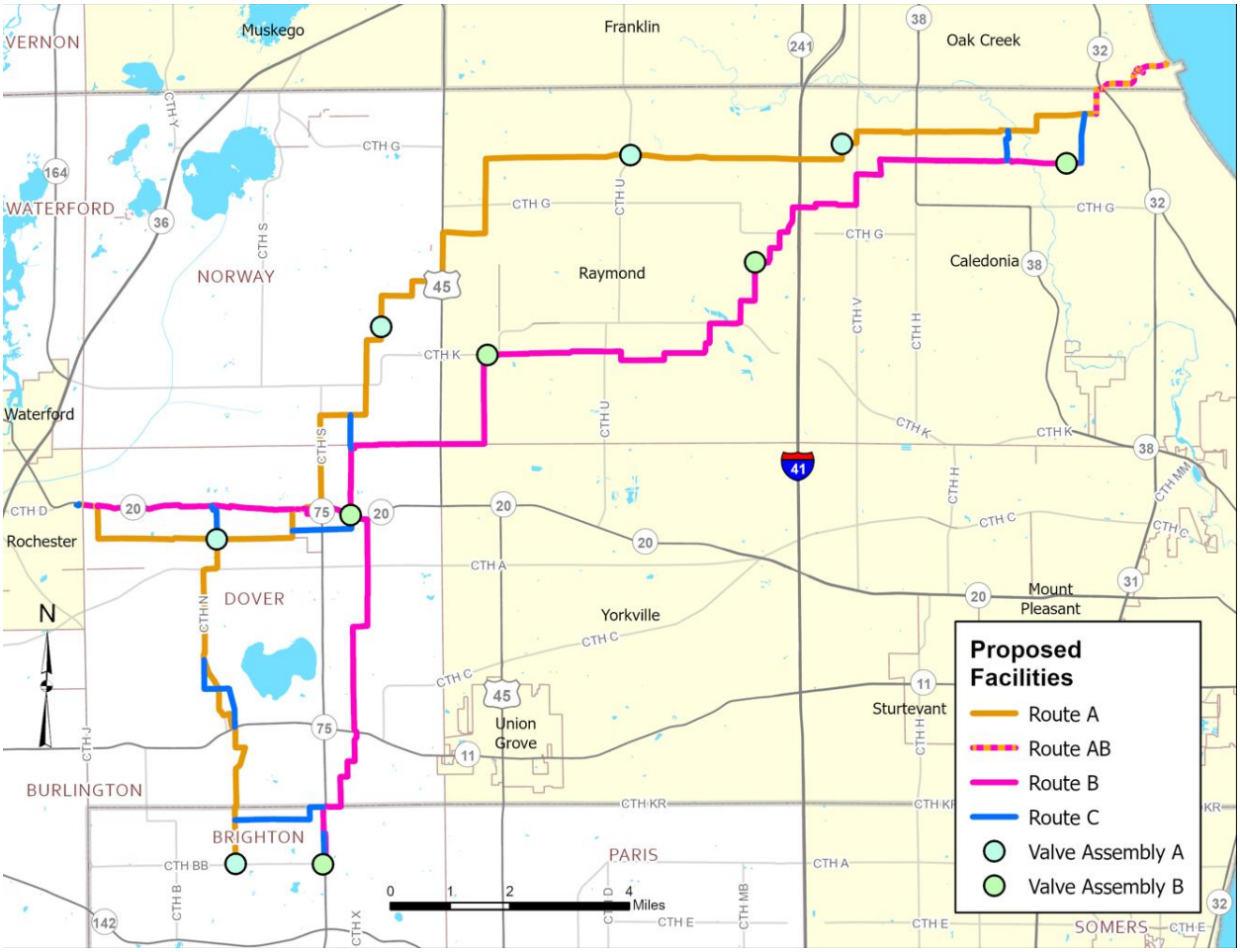
3.1. Route Location

The RLP would be new construction that would tie supply and load together from the existing WE-GO Bluff Creek Liquified Natural Gas Facility in the town of LaGrange to the combined Oak Creek and Elm Road Generating Station campus in the city of Oak Creek by connecting the Lakeshore Lateral Project in the town of Dover to the Rochester ANR Gate Station in the village of Rochester.

Proposed Route A starts at the existing Rochester Gate Station and continues east. The southern portion of Route A connects to the Lakeshore Lateral at 7th Street, abutting Bong State Recreational Area in Brighton to the north, and follows Sunnyside Drive north until connecting with the section of Route A that started from Rochester Gate. From there, Route A follows North Beaumont Avenue north, eventually crosses over Raynor Avenue and follows agricultural fields to the east. Route A crosses US Hwy 41 and the Root River Parkway until connecting with the Oak Creek campus located on the shores of Lake Michigan. Proposed Route B starts at the existing Rochester Gate Station, largely follows Hwy 20, and crosses over agricultural fields. From the south, Route B starts at the Lakeshore Lateral on Hwy 75, which abuts Bong State

Recreational Area on the east. Route B goes through agricultural fields, crosses over Church Road and Plank Road as it continues north and connects with Route B from the Rochester Gate at Hwy 20 in Beaumont. From there, it continues north and east, crossing North Raynor Avenue. It continues northeast, crossing roads such as County Road K, 5½ Mile Road, US Hwy 41, County Road V, and Hwy 38. As it continues east, is crosses a section of the Root River, heads north across 7 Mile Road, and meets its destination at the Oak Creek campus. Figure 1 illustrates the locations of Routes A, B, and C.

Figure 1: Project Overview



Routes A and B are both similar lengths. Route A is about 32.9 miles and Route B is about 33.2 miles. The majority of either route would be located on agricultural land, followed by grassland, as shown on Table 1 below[2]. A table showing temporary and permanent impacts per land use category can be found in a data request response filed under PSC REF#: 538345[3]. The applicant stated that permanent and temporary impacts are reflected by easement type. All above ground facilities (such as the new valve assemblies and expanded Rochester Gate Station) are

[2] PSC REF#: 525537 Appendix F Attachment 2 PSC Table 2 – Land Cover
[3] PSC REF#: 538345 Response-Data Request-PSC-Ingwell-VBL-03242025 Attach 01

accounted for the in Table 2 – Land Cover of the application.[4,5] Other above ground facilities (such as the SCCT meter set and high-pressure regulator station) are located on private company property.

[4] PSC REF#: 525537 Appendix F_Att 02_PSC Table 2-Land Cover
[5] PSC REF#: 528953 Response-Data Request-PSC-Thompson-EAT-2.03

Table 1: Land Use Per Route

Route	Agricultural	Grassland	Non-Forested Wetland	Forested Wetland	Upland Forest	Developed/Urban	Totals	Route + Staging areas total
Route A	239.1 acres/ 61.7 percent	64.4 acres/ 16.6 percent	29.5 acres/ 7.6 percent	5.7 acres/ 1.5 percent	20.7/ 5.3 percent	28 acres/7.2 percent	387.4 acres	521.4 acres
Route A Staging Areas	30.3 acres	57 acres	1.9 acres	0 acres	0 acres	44.8 acres	134 acres	
Route B	246.4 acres/62.5 percent	61.1 acres/15.5 percent	30.6 acres/7.8 percent	4.9 acres/1.2 percent	24.6 acres/6.2 percent	26.8 acres/6.8 percent	394.4 acres	528.4 acres
Route B Staging Areas	30.3 acres	57 acres	1.9 acres	0 acres	0 acres	44.8 acres	134 acres	
Segment C-1	0.1 acres	0.4 acres	0 acres	0 acres	0 acres	0 acres	0.5 acres	223.6 acres
Segment C-2	6.9 acres	0.5 acres	0 acres	0 acres	0 acres	0 acres	7.4 acres	
Segment C-3	10.5 acres	0.1 acres	3 acres	0 acres	0.4 acres	0.1 acres	14.1	
Segment C-4	0 acres	0.2 acres	0 acres	0 acres	0 acres	1.2 acres	1.4	
Segment C-5	5.8 acres	0.5 acres	0.4 acres	0 acres	0 acres	0 acres	6.7	
Segment C-6	4.5 acres	0.8 acres	0.3 acres	0 acres	0.3 acres	0.1 acres	6	
Segment C-7	6.6 acres	0.6 acres	0.8 acres	0.3 acres	1.6 acres	0.1 acres	10	
Segment C-8	8.1 acres	0 acres	0.6 acres	0 acres	0 acres	0 acres	8.7	
Segment C-9	7.7 acres	0.2 acres	2 acres	0 acres	0 acres	0.1 acres	10	
Segment C-10	16.9 acres	0.8 acres	1.8 acres	0.1 acres	0 acres	0 acres	19.6	
Segment C-11	2 acres	1.8 acres	0.6 acres	0.4 acres	0.3 acres	0.1 acres	5.2	
Route C Staging Areas	30.3 acres	57 acres	1.9 acres	0 acres	0 acres	44.8 acres	134 acres	

WE-GO divided the construction of Routes A, B, and C into segments. "A" segments are segments specific to Route A. "B" segments are specific to Route B. "AB" segments are segments where only one feasible route was identified. "C" segments are segments that connect Route A and Route B at various locations that would allow for crossover between Route A and Route B segments. Route A consists of 19 segments, Route B consists of 18 segments, and Route C consists of 11 segments. Appendix A Attachment 12 of the application[6] describes the details of each segment.

Approximately 151,316 feet of proposed easement would be used for Route A and 155,767 feet of proposed easement would be used for Route B. The typical proposed permanent easement width is 50 feet.

3.1.1 Minor Route Adjustments

Under the Commission's typical standard order conditions, if the project is approved, the applicant would be allowed to propose minor adjustments to the approved route for the protection of environmental resources, to respond to landowner requests, or to implement technical design changes that arise during the final stages of engineering. However, the applicant would not be allowed to propose any changes from the approved pipe location that would affect resources not discussed in the EA, nor any changes that would affect new landowners who had not been given proper notice and hearing opportunity or landowners who had been given proper notice and hearing opportunity in a significantly different manner than was originally approved, nor any changes that include a unique occurrence not discussed in the EA of, for example, a particular human burial, archaeological site, or protected species. The applicant would be required to consult with Commission staff regarding whether the change rises to the level where Commission review and approval is appropriate. For each proposed adjustment for which Commission review is appropriate, the applicant would be required to submit for Commission staff review and approval a letter describing:

1. the nature of the requested change;
2. the reason for the requested change;
3. the incremental difference in any environmental impacts caused by the change;
4. communications with all potentially affected landowners regarding the change;
5. and a map showing the approved route and the proposed modification, property boundaries, and relevant natural features such as woodlands, wetlands, waterways, and other sensitive areas.

Documentation would be required to include all the information provided to the landowner regarding the changes, include any feedback provided by the landowner, identify any way in which landowner feedback has informed the changes proposed, and whether the landowner agrees to the proposed changes.

[6] PSC REF#: 519005 Appendix A Attachment 12 Project Route Detailed Description

Approval of the minor route adjustment request is delegated to the Administrator of the Division of Energy Regulation and Analysis with advice and consent from the Administrator of the Division of Digital Access, Consumer, and Environmental Affairs.

4. Project Construction

If the proposed project is approved, WE-GO must obtain, in addition to a Commission issued Certificate of Authority, all applicable state, local, or municipal permits prior to initiating construction. Construction is planned to start in late 2026, with mobilization, clearing and brush removal, grading and soil segregation conducted in fall 2026. Pipe sections would be delivered and positioned along the route at this time also. Main installation would take place through the majority of 2027, with anticipated completion by fall 2027. Restoration activities may occur simultaneously with main installation activities.

4.1 Construction Methods and Design

The transmission main would be installed by a combination of open trench, horizontal directional drilling (HDD) and jack and bore operations. Open trenching is a commonly used type of construction in which excavated materials are replaced after a trench has been dug and the pipe has been installed. A typical trench excavated for this project would measure, on average, eight feet wide by seven feet deep for 24-inch and 30-inch pipe. Individual sections of pipe with a factory-applied epoxy coating are laid in the ROW (a process known as pipe stringing) and bent as necessary to accommodate the contours of the trench. The supported pipe sections are welded by qualified welders and all welds are visually and radiographically inspected by certified inspectors and technicians. All final welds are field coated to deter corrosion, and the coating is tested for any voids in a process commonly known as "jeeping." After the entire pipe section is inspected to ensure that the corrosion coating is adequate and no dents exist, the pipe is lowered into the trench using side tractor booms and the trench is backfilled with subsoils. Subsoils should be kept from mixing with stockpiled topsoil. Prior to placing the new pipeline into service, hydrostatic testing would occur to ensure the integrity of the pipe. Water would be pumped into sections and held at design pressures for a minimum of eight hours to check for leaks. A calipered inline inspection tool would also be sent through the pipe to detect dents or weaknesses in the pipe.

In areas where the soil has limited cohesion, the trench width may need to be widened to ensure that adequate depth of cover for the gas pipe is achieved. In agricultural lands, trench depth would be sufficiently deep enough to allow a minimum of four feet of cover over the top of the pipeline to avoid possible interference with farming equipment. Material excavated during trenching would be temporarily piled to one side of the ROW or acquired easement, with topsoil and subsoil separated, if applicable. After the pipe is placed in the trench, topsoils are replaced at the surface. In particularly rocky soils, some additional rock-free fill material may be brought in to pad the trench bottom and the pipe during the backfilling process. After proper filling, the replaced materials are compacted to minimize the potential for future settling.

Site cleanup and restoration would begin as soon as soil and weather conditions permit. Construction trash, debris, and remaining large rocks would be removed from the construction area. The original soil contours and tilt would be restored through grading and decompaction as needed, and disturbed areas would be reseeded. The choice of seed mixture is often made based on land-use and adjacent properties, with some consideration given to landowner preferences. Agricultural fields that would be seeded with crops are often not reseeded by the utility with

permanent vegetation. Erosion control measures would be maintained until the soils are stabilized or the site is re-vegetated. Fences that were previously cut or removed would be replaced, and pipeline markers and mainline valves would be placed along the ROW as needed.

4.1.1 Construction methods for crossing sensitive resources or man-made features

HDD is a technique that is used to cross environmentally sensitive areas and where minimizing excavation disturbance is crucial. The feasibility of HDD is largely dependent on the local geologic setting. WE-GO would conduct geotechnical studies where necessary to confirm that conditions at the anticipated HDD crossings are suitable for this technique. During HDD construction, an entry and exit pit are excavated on either side of the feature being avoided (e.g., a high-quality stream) and a small diameter pilot hole is drilled beneath the feature using electromagnetic sensors to guide the drill. Dewatering of bore pits may be required. Additional temporary workspace requirements, usually 100 by 200 feet, are needed at both ends of the area being drilled to set up the drilling machine and associated equipment. The size of those workspaces is determined by pipe diameter and local conditions. The pilot hole is then enlarged to the necessary diameter using reaming tools that rotate at high speed and are drawn back toward the drilling rig. During this process, a pressurized drilling fluid, composed of water, bentonite clay, and synthetic polymers is pumped into the hole to lubricate the reaming tool, remove soil cuttings, and maintain the integrity of the hole. Bentonite is an inert, non-toxic clay that lubricates and supports the hole and can form a sealed layer of clay around the borehole. When the hole is the appropriate size, the welded pipe is attached to the reaming tool at the exit side and pulled through the hole towards the drill rig.

Used drilling mud is often stored temporarily on-site using containment pits or berms to prevent the material from flowing into streams, wetlands or other sensitive environments. After the crossing is complete, the material is disposed of at an appropriate upland site or disposed of in accordance with applicable permits or regulations.

Although this method is costlier than open trenching, it requires less ground surface disturbance, thereby reducing the potential risk associated impacts, including soil loss and introduction of invasive species. Environmental risks associated with HDD include the escape of drilling mud into the environment as a result of a spill, tunnel collapse, or the rupture of mud to the surface, commonly known as an inadvertent release or frac-out. A frac-out is caused when excessive drilling pressure results in drilling mud moving toward the surface. It can happen if the drilling fluid finds a crack, fissure, or loose material where the fluid more easily passes through to the surface. If this occurs within a waterway, the drilling mud can be carried downstream and/or distributed along the stream bottom decreasing water quality and destroying habitat. Although the bentonite clay is non-toxic, the fine clay particles can smother plants and animals in an aquatic environment. To minimize any adverse effects associated with frac-outs, the following points would be incorporated into the HDD process:

- The bore path would be monitored for evidence of a frac-out for prompt response.

- Appropriate containment materials would be on-site such as silt fence, hay bales and sandbags.

- Vacuum excavation would be readily available on short notice to respond quickly to a frac-out.

WE-GO provided a copy of its response plan for the inadvertent release of drilling lubricants during HDD activities which describes the responses to be undertaken should a frac-out occur.

The jack and bore method would be used to for railway crossings. Bore pits are excavated on each side of the obstruction that would be crossed. The entrance and exit bore pits begin and end within the pipe trench and range in depth from 12 to 15 feet depending on the depth of cover required for the pipe being installed. If groundwater is encountered in the bore pits, it would be pumped from the pits into a dewatering structure. Access pads (such as timber or heavy plywood) may be placed to allow heavy tracked equipment to cross the roadway without damaging the paved surface. The auger boring machine is set up in the entrance pit and a casing pipe is jacked under the obstruction while the earth is removed by the auger rotating inside the casing pipe. The new carrier pipe is attached to the casing pipe and is either pushed or pulled under the road or railway. After the new carrier pipe is installed and tied into the rest of the pipeline, the bore pits would be backfilled with the stockpiled soil. Access pads and dewatering structures would be removed, and the stockpiled topsoil would be replaced to the original grade. Temporary erosion control measures would be removed after permanent erosion control measures are installed and vegetation is re-established.

Construction equipment used during the project would include dozers, graders, excavators, trenchers, dump trucks, back hoes, side booms, horizontal directional drill rigs, pickup trucks, vacuum excavators, rippers, tillers, rock picking machines, welding rigs and trucks, and x-ray trucks.

4.2 Use of Public Road ROW and Private Easement Acquisition

If the project is approved, it would be constructed on both private easements and public road ROW. In areas where the route would cross private lands, WE-GO would negotiate with landowners prior to construction to secure the necessary easements. For instance, in agricultural areas, a 50-foot permanent easement and a 50-foot temporary construction easement would be acquired, while in non-agricultural lands adjacent to road ROW, a maximum 50-foot permanent easement paired with a 25-foot temporary easement would be used—with the non-paved ROW serving as temporary workspace. Typically, for large natural gas pipelines (such as transmission lines or pipelines greater than 12 inches in diameter), the ROW consists of a permanent easement within which the pipe resides and additional temporary easements that provide the workspace needed for construction activities, including space for equipment and stockpiled soils removed from the trench. Often, pipelines are installed on private property to avoid conflicts with existing utilities and future road projects. However, when the pipeline must be located within road ROW to avoid impacts on houses or other resources, coordination with WisDOT or other road authorities—and obtaining the necessary permits—is essential. Generally, the construction disturbance may extend 75 to 100 feet in width, with the core 50-foot permanent easement supplemented by a temporary easement (25 to 50 feet) that is released upon project completion.

In some areas, alternative construction techniques may further reduce the impacted ROW, although these methods can require larger temporary workspaces at the ends of the crossing areas.

4.3 Off-ROW Requirements

Off-ROW access paths and/or roads are occasionally needed where access is restricted due to steep slopes, waterway crossings, railroads, or other access limitations. These are typically 30-feet wide to allow for a range of construction traffic. Utilities often seek to use existing farm tracks or lanes if working in agricultural settings, or other types of informal roads. If necessary, a new off-ROW access road could be constructed. Prior to construction, off-ROW access roads may need modifications and improvements to allow for safe equipment movement to and from the ROW. These modifications may include vegetation removal, grading, and/or gravel placement.

Additional off-ROW areas that would be utilized as a part of the proposed project include laydown yards, which are also commonly referred to as staging areas. These areas are used for storing construction materials, vehicles, temporary staff buildings, and structures. Suitable locations for laydown yards/staging areas are clear, open, generally flat areas near or along the project ROW where materials and equipment can be delivered and stored until needed during construction. Unused parking lots, fallow fields, or brownfields are commonly used for staging areas. Prior to use, these areas are screened via desktop resources and field-investigated to discern the presence or absence of sensitive natural resources, wetlands and waterbodies, and cultural resources that may need to be fenced off or avoided. If necessary, minor grading might occur on the site and aggregate materials would be brought in to provide level surfaces for temporary buildings, vehicle parking, and storage. Agreements with the landowner(s) for these areas (when applicable) would specify how any specific laydown yard, staging area, or off-ROW access road would be restored or left after work was completed.

4.4 Restoration of Impacted Properties and ROW

WE-GO would restore road ROW and acquired easements to preconstruction conditions where permissible. Surface grading would be done to re-establish natural contours. Decompaction of soil would occur where necessary and re-vegetation would be compatible with preconstruction conditions and adjacent vegetation patterns.

4.5 Long-term management of the natural gas pipeline ROW

Unlike electric transmission line ROWs, which are often maintained as an 80- to 150-foot zone clear of all tall woody vegetation and may even require trimming or removal of danger trees outside of the ROW, the long-term maintenance of pipeline ROWs is not as stringent. Ultimately, the vegetation must be maintained in a state that allows aerial inspection of the pipeline corridor and an accessible path, up to 25 feet wide and centered over the pipeline, to enable periodic inspections on the ground.

Cropland, as well as grassland or wetland areas supporting non-woody vegetation, likely would not require any periodic mowing or form of long-term maintenance. Similarly, grassy areas with low-growing scattered shrubs or trees may not need to be formally maintained.

In more densely forested areas or those with a heavy shrub understory, the need to maintain access for aerial and ground inspection could result in trees and shrubs being cut as needed. Many utilities manage a ROW where trees within 10 to 15 feet of the pipe and greater than 15 feet in height may be cut and removed. In areas where the pipeline may have been horizontally directionally drilled beneath a forested area, accessibility to the pipeline corridor would still be necessary to allow ground inspections, but this may only need to be wide enough to allow a footpath or ATV.

4.6 Permits and Approvals Required

WE-GO would require the following permits for the project and would continue to work with local units of government to identify any additional permits that may be required.

Table 2: List of Permits Required

Agency	Permit Name/Permit Subject
Wisconsin Department of Natural Resources (DNR)	Construction Site Erosion Control and Storm Water Discharge General Permit
Wisconsin Department of Natural Resources (DNR)	Hydrostatic Test Water General Permit
Wisconsin Department of Natural Resources (DNR)	Wetland Disturbance General Permit
Wisconsin Department of Natural Resources (DNR)	Chapter 30 Dredging General Permit
Wisconsin Department of Natural Resources (DNR)	Chapter 30 Grading General Permit
Wisconsin Department of Natural Resources (DNR)	Chapter 30 Temporary Clear Span Bridge General Permit
Wisconsin Department of Natural Resources (DNR)	Chapter 30 Miscellaneous Structure General Permit
Wisconsin Department of Natural Resources (DNR)	Incidental Take Authorization/Permit (TBD)
Wisconsin Department of Natural Resources (DNR)	High Capacity Well Permit
Wisconsin Historical Society	Permission to Disturb Authorization
US Army Corps of Engineers	Utility Regional General Permit
US Fish and Wildlife	Incidental Take Authorization/Permit or Consultation Requirements
Wisconsin Department of Transportation (WisDOT)	Utility in ROW
Kenosha County	Utility in ROW
Racine County	Utility in ROW
Town of Brighton	Utility in ROW
Town of Dover	Utility in ROW
Town of Norway	Utility in ROW
Village of Caledonia	Utility in ROW
Village of Caledonia	Erosion Control Permit
Village of Raymond	Utility in ROW
Village of Rochester	Utility in ROW
Racine County and/or Village of Rochester	Driveway/Roadway Permit, if required (one for each VA & Rochester Gate)
Racine County and/or Village of Rochester	Zoning Permit, if required (one for each VA & Rochester Gate)
Racine County and/or Village of Rochester	Conditional Use Permit, if required (Rochester Gate)
Racine County and/or Village of Rochester	Erosion Control Permit, if required (Rochester Gate)
Racine County and/or Village of Rochester	Building Permit(s), if required (Rochester Gate)
Milwaukee County and/or City of Oak Creek	Conditional Use Permit, if required (HPDR)
Milwaukee County and/or City of Oak Creek	Zoning Permit, if required (HPDR)
Milwaukee County and/or City of Oak Creek	Driveway/Roadway Permit, if required (HPDR)
Milwaukee County and/or City of Oak Creek	Erosion Control Permit, if required (HPDR)
Milwaukee County and/or City of Oak Creek	Building Permit(s), if required (HPDR)
Village of Rochester and/or Racine County – Route A	Other permitting requirements, driveway, fence, zoning, etc TBD (Valve Assembly)
Town of Dover and/or Racine County – Routes A&B	Other permitting requirements, driveway, fence, zoning, etc TBD (VA & Receiver)
Town of Brighton and/or Kenosha County – Routes A&B	Other permitting requirements, driveway, fence, zoning, etc TBD (VA & Launcher)
Town of Norway and/or Racine County – Route A	Other permitting requirements, driveway, fence, zoning, etc TBD (VAs)
Village of Raymond and/or Racine County – Routes A&B	Other permitting requirements, driveway, fence, zoning, etc TBD (VAs)
Village of Caledonia and/or Racine County – Route A&B	Other permitting requirements, driveway, fence, zoning, etc TBD (VAs)
Racine County	Zoning permit, if required (one for each staging area)
Village of Union Grove	Zoning permit, if required (staging area)
Town of Waterford	Zoning permit, if required (staging area)
Village of Raymond	Zoning permit, if required (staging area)
City of Franklin	Zoning permit, if required (staging area)
Milwaukee County	Zoning permit, if required (one for each staging area)
Village of Yorkville	Zoning permit, if required (staging area)
City of Oak Creek	Zoning permit, if required (staging area)
Village of Caledonia	Zoning permit, if required (staging area)
Town of Norway	Zoning permit, if required (staging area)
Town of Dover	Zoning permit, if required (staging area)
Village of Sturtevant	Zoning permit, if required (staging area)

4.7 Staging Areas

It is anticipated that up to four staging areas would be selected from the proposed 17, consisting of approximately one to 13 acres at each location. Most of the proposed staging areas are comprised of developed lands including electric facilities, commercial operations, parking lots, a mowed field, and an agricultural field. WE-GO states in its application that it intends on selecting staging area sites where no adverse environmental impacts would occur. Prior to final selections, field determinations and delineations of environmental features would be completed with the intent to avoid potential impacts.

5. Environmental Impact Analysis

The proposed project is a Type II action under Wis. Admin. Code § PSC 4.10(2). The Division of Digital Access, Consumer, and Environmental Affairs of the Commission has prepared this EA to determine if an EIS is necessary under Wis. Stat. § 1.11. A notification of the Commission's intent to prepare an EA, which included a solicitation for comments on the environmental aspects of this project, was mailed to the entire project mailing list on November 20, 2024. The main issues voiced in the responses included concerns about gas leaks/explosions, damage to wildlife and ecosystems, and proximity of the pipeline to residences. These responses are stated in detail in Section 1.3.2 of this EA.

5.1 Forested Lands

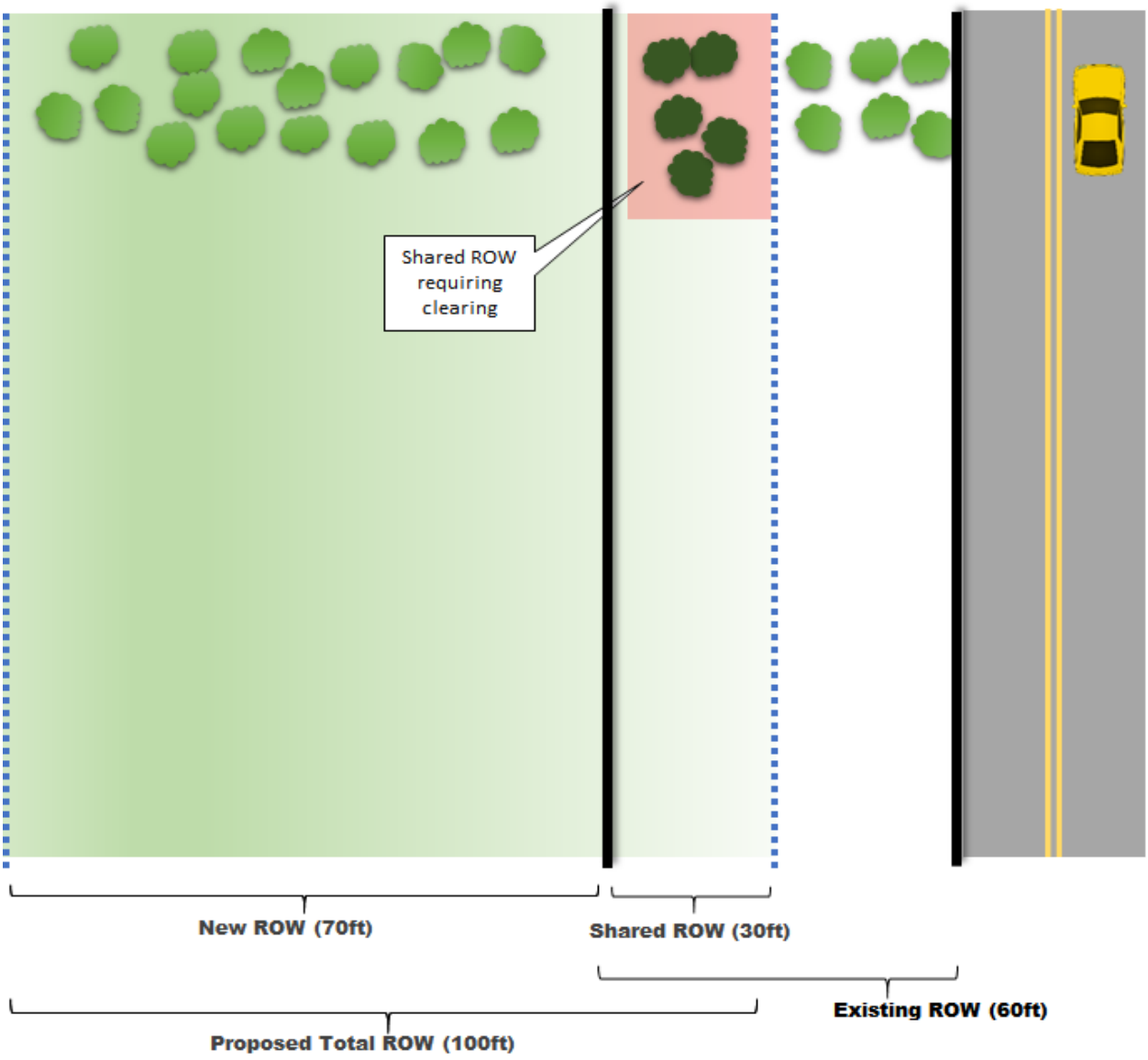
Forested lands were identified and reviewed by the company via desktop analysis and observations from fieldwork completed in 2024 where access was permitted or from adjacent accessible ROWs and parcels. Excluding staging areas, approximately 20.7 acres of upland forested land are located along proposed Route A, 24.6 acres are located along proposed Route B, and 2.5 acres are located along proposed Route C. According to the applicant's Land Cover Table,[7] the following number of acres per route are located in shared versus new ROWs. A picture depicting ROW terms is shown in Figure 3 below.

Table 3: Forested Land Impacts Per Route

Route	Shared ROW (acres)	Shared ROW Requiring Clearing (acres)	New ROW (acres)	Total acres of forest impacted	Permanent impacts (shared + new ROW acres)
Route A	6.1	5.9	14.6	20.7	10.4
Route B	3.9	3.7	20.7	24.6	16
Route C	0.3	0.2	2.2	2.5	1.7

[7] PSC REF#: 525537 Appendix F Attachment 2 PSC Table 2 – Land Cover

Figure 2: ROW Example



Additionally, 5.7 acres of forested wetland are located along proposed Route A, 4.9 acres are located along proposed Route B, and 0.9 acres are located along proposed Route C. No upland forests are located within the staging areas, however, there is <0.1 acre of forested wetland, but this is planned to be avoided during construction. Most forests along the project are privately owned and likely used for recreation. Four segments (A-7, A-9, B-7, and AB-3) run through portions of forested public land.[8] Segment A-9 abuts Bong State Recreational Area on the north and Segment B-7 runs through a Racine County-owned forest and wetland area which includes the Root River. Segment A-7 also crosses a forested section of the Root River.

[8] PSC REF#: 519137 Appendix F Attachment 24 Forestry Table

Forested lands along Routes A, B, and C generally consist of fragmented wooded areas associated with cropland, residential properties, and/or waterways. Larger blocks are predominantly mesic to dry-mesic oak-hickory communities comprised of a mix of boxelder (*Acer negundo*), red and/or maples (*Acer rubrum, A. saccharum*), black cherry (*Prunus serotina*), red, white, and/or bur oaks (*Quercus rubra, Q. alba, Q. macrocarpa*), shagbark hickory (*Carya ovata*), and some American Elm (*Ulmus Americana*) in the canopy with common buckthorn (*Rhamnus cathartica*) and invasive honeysuckle shrubs (*Lonicera spp.)* prevalent throughout. Smaller and/or more degraded forested blocks generally have a greater component of black walnut (*Juglans nigra*) and boxelder within the canopy and a higher occurrence of invasive species. Areas of forested wetland include mostly hardwood swamp communities. Common canopy species within forested wetlands included silver maple (*Acer saccharinum*), green ash (*Fraxinus pennsylvanica*), red maple, and boxelder with some occasional willows (*Salix spp.).* Common buckthorn was pervasive within the understory of forested wetlands. Most forests along the project are privately owned and likely used for recreation. The predominant tree sizes ranged from saplings to medium-sized (20-30 inches diameter at breast height (dbh)), with instances of large trees (greater than 30 inches dbh) (typically white oaks) occasionally encountered in the larger forest tracts.

There are no Forest Crop Law (FCL) lands along the route segments for this project. Seven properties enrolled in the Managed Forest Law (MFL) are intersected by or directly adjacent to the project area, three by Route A and four by Route B. Of these properties, two on Segment A-6 are anticipated to require clearing, totaling 2.3 acres of forest cleared.[9] The other five properties are either currently utilized in active agriculture or have forested lands that end just at the border of the easement area and would require no or minimal branch overhang clearing.

Trees and shrubs would be cut as needed within the project ROW to create the required workspace prior to construction. Regular maintenance clearing of trees and shrubs over natural gas lines is also necessary to facilitate periodic inspections and surveys. A corridor along the pipeline typically 20 feet wide may be maintained in an herbaceous state (cleared of trees and shrubs) within uplands as needed. The maintained cleared path would be no greater than 20 feet wide in forested wetland communities. Trees and brush would be cleared for the full width of the ROW to facilitate construction equipment access and ensure safe clearances between vegetation and the pipeline. Vegetation would be cut at or slightly above the ground surface. Rootstock would generally be left in place except in areas where stump grinding is necessary to facilitate the movement of construction vehicles, or if requested by the landowner. As part of the initial clearing, equipment such as mechanized mowers, sky trims, processors, and harvesters would be used. Dozers, excavators, trenchers, dump trucks, vacuum excavators, rippers, tillers, rock picking machines, welding rigs, pickup trucks, and x-ray trucks may be used during the installation of the main and final restoration.

Forested land clearing activities would take place during fall 2026 and winter 2027, pending project approval. The timeframe for clearing activities in certain areas may be adjusted to comply with any restrictions pertaining to specific wildlife and/or vegetation presence, such as

[9] PSC REF#: 525832 Response-Data Request-PSC-Thompson-EAT-1.59

birds and bats. The requirements for rare bird species would be followed, which involves avoidance of suitable habitat within one mile of the known rare bird occurrence or conducting surveys to determine presence. There are no known occurrences of bats or bat hibernaculum, and there are no known roost trees present within the vicinity of the project corridor. The company would work with US Fish and Wildlife Service (USFWS), as necessary, through the Army Corps of Engineers wetland permitting process to ensure federal species compliance.

Construction impacts in and around forested lands would be minimized by reducing the extent of the clearing area, utilizing HDD under forested wetland areas as feasible, and by using appropriate post-construction seeding and management to reduce the spread of invasive species and encourage re-establishment of vegetative cover.

5.2. Grasslands

Grasslands were identified and reviewed via desktop analysis and observations from field surveys completed in 2024 where access was permitted or from adjacent accessible ROWs and parcels.

The most observed grassland areas were those in roadsides or adjacent roadside and transmission line ROWs. These grassland areas are typically dominated by exotic cool season grasses such as smooth brome (*Bromus inermis*) and Kentucky blue grass (*Poa pratensis*) with a variety of exotic/invasive and native forbs including wild parsnip (*Pastinaca sativa*), Canada thistle (*Cirsium arvensis*), dandelion (*Taraxacum officinale*), Queen Anne's-lace (*Daucus carota*), Canada goldenrod (*Solidago canadensis*), various clovers (*Trifolium spp*.), bird's foot trefoil (*Lotus corniculatus*), and common milkweed (*Asclepias syriaca*). In addition to roadside and transmission ROW grasslands, old fields, pastures, hayfields, as well as vegetated drainage swales, fence lines, or field lines associated with agricultural lands and residential properties were observed. These grasslands contained similar species as the roadside grasslands, in addition to sweet clover (*Melilotus alba*), tall fescue (*Lolium arundinaceum*), orchard grass (*Dactylis glomerata*), timothy (*Phleum pratense*), common plantain (*Plantago major*), and scattered woody species. The hayfields in particular contained a mix of alfalfa, orchard grass, Kentucky bluegrass, tall fescue, red clover, and other common agricultural weeds.

Excluding staging areas, approximately 64.4 acres of grassland are located along proposed Route A, 61.1 acres of grassland are located along proposed Route B, and six acres of grassland are located along proposed Route C. The proposed staging areas contain 57.0 acres of grassland. There would be 27.1 acres of permanent impacts along Route A, 24.7 acres along Route B, and 3.6 acres along Route C. Roadside grasslands along the project are publicly owned, routinely maintained natural areas. Old field, hayfields, and inactive pasture grassland areas along the project are privately owned and used for recreation, non-productive agricultural, and/or natural area uses.

The applicant stated that vegetation disturbance within grasslands would be temporary in nature. Soil disturbance would be limited to the trenches and the pits necessary for the pipeline and HDD. Precautionary protocols during construction and post-construction seeding and

management are expected to reduce the spread of invasive species and encourage the re-establishment of vegetative cover. Many of the grassland areas identified during fieldwork are comprised of roadside herbaceous communities that would be seeded with a standard Wisconsin Department of Transportation (WisDOT) seed mix. Natural areas or prairie grasslands with a variety of species may be seeded with a native seed mix once construction is complete.

Several additional measures can be taken by the applicant to avoid or mitigate impacts to grassland resources:

- Avoid placing structures and spoils where sensitive plant species are present and avoid sensitive grasslands when possible.
- Use low-pressure tires on construction equipment to minimize ground disturbance in prairie areas.
- Use timber matting to reduce impacts to the ground surface.
- Implement erosion control BMPs to reduce potential topsoil loss and storm water runoff.
- When possible, develop access paths within the ROW to minimize soil disturbance and impacts to existing vegetation.
- Implement time of year restrictions. Since ROW clearing during the growing season is extremely impactful to grassland resources, this impact could be mitigated by clearing during the winter months when vegetation is dormant.

5.3 Federal Emergency Management Agency (FEMA) Flood Hazard Areas

FEMA floodplains or flood-prone areas are depicted on the FEMA Flood Hazard Map included in Appendix F Attachment 5[10] of the WE-GO application**.** Flood hazard areas were derived from the FEMA National Flood Hazard Layer (NFHL). There are 14 flood hazard areas mapped on Route A, seven areas mapped on Route B, and five areas on Route C. Additionally, there is one flood hazard area mapped within staging area 11. Flood hazards include 100-year Flood Zones and Floodways and 500-year Flood Zones. Floodplain impacts would be temporary, and disturbances would be restored to grade. Relevant floodplain authorities and local floodplain zoning ordinance permits would be obtained prior to construction as necessary.

5.4 Wetlands

Wetlands provide vital functions that benefit society. Wetlands detain storm water runoff, enabling the slow recharge of groundwater resources and lowering downstream peak flood levels; filter sediments and pollutants from the air, precipitation, and upstream sources to support higher water quality downstream; provide food, cover, and nesting habitat for many species of fish and wildlife; provide recreational opportunities for bird watching, hunting, hiking, and enjoyment of

[10] PSC REF#: 525530 and 525539 Att 05 Flood Hazard Map (1) & (2)

the surrounding landscape. It is estimated between one-quarter and one-third of all rare species in Wisconsin are found in wetlands.

Wetlands are a dynamic ecosystem and provide different functions depending on the type of wetland. The same wetland may even provide different functions from year to year and season to season. There are many different types of wetlands, typically characterized by the size, type of vegetation, and amount of soil saturation and/or surface water found within them.

Standardized assessment methods are used to evaluate the extent to which a specific wetland may perform any given function. The presence or absence of specific characteristics is used to determine the importance of each functional value for the site in question. Wetland functional values are determined by assessing multiple physical, chemical, and biological parameters, such as floral diversity, fish and wildlife habitat, flood protection, water quality protection, shoreline protection, groundwater recharge and discharge, recreation, and aesthetics.

5.4.1 Wetland Identification and Quality

5.4.1.1 Wetland Identification Methods

Wetland presence was evaluated for Routes A, B, C, access roads, and staging areas through a combination of field determinations, refined desktop determinations, and one previous wetland determination from another project.

Field and desktop determinations were conducted within a survey corridor that included, but extended beyond the boundaries of, the proposed Routes A, B, C, access roads, and staging areas; collectively, this survey area is termed as the "Project Survey Area."

Wetland determinations within the Project Survey Area were performed during the growing season between April and July 2024 and generally following the criteria and methodology described in the United States Army Corps of Engineers (USACE) *Corps of Engineers Wetlands Delineation Manual, Technical Report Y-87-1 (1987)* and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Midwest Region*.

Wetland determinations included offsite evaluation of Google Earth aerial imagery, U.S. Department of Agriculture Farm Service Agency National Agriculture Imagery Program (NAIP) aerial imagery, Natural Resources Conservation Service (NRCS) soil survey data, DNR LiDAR mapping data, DNR Wisconsin Wetland Inventory (WWI), DNR Wetland Indicators and Soil data, and DNR 24K Hydrography data.

Field determinations were conducted within the Project Survey Area where the applicant either had direct access to an area (termed "field determined" by the applicant) or visually assessed from a distance where the applicant had access permissions, such as road ROW (termed "field verified" by the applicant). Soil samples were generally not performed during the field determinations and the applicant relied on a combination of mapped NRCS soil data and professional judgement to evaluate the presence or lack of hydric soil indicators.

A previously reviewed and approved wetland determination report for the PSC-regulated Oak Creek CT Project (PSC Docket 6630-CE-317) was also applied to the Project Survey Corridor on the far eastern portion of the project, east of HWY 32 near the Oak Creek Power Plant property. DNR provided wetland consultation on the Oak Creek CT wetland determination report in March 2024.

Without field survey data, the presence, size, plant community, and/or quality of wetlands within the Project Survey Area cannot be confirmed. The applicant stated additional field surveys would occur in 2025 on properties not yet surveyed and where landowners provide access consent. The applicant also stated that prior to construction, all staging areas and areas of permanent fill would be field visited to evaluate for the presence of wetland.

5.4.2.1 Wetland Identification

Route A

Approximately 174 wetlands, totaling 35.5 acres, were identified for Route A. Wetland types were identified as 1.5 acres of Floodplain Forest, 13.5 acres of Fresh (Wet Meadow), 3.0 acres of Hardwood Swamp, 12.9 acres of Seasonally Flooded Basin, 0.2 acres of Sedge Meadow, 2.8 acres of Shallow Marsh, and 1.5 acres of Shrub-Carr. Wetland community types were identified as 4.5 acres of forested wetland, 1.5 acres scrub-shrub wetland, and 29.5 acres emergent wetland.

The applicant further categorized the wetland types as either degraded or not degraded. Generally, the applicant describes “degraded” wetlands as wetlands that may provide lower functional values and be lower in quality. Examples include wetlands identified as constructed drainage features or farmed depression, wetlands with low floristic diversity, and wetlands dominated by invasive species. Approximately 5.9 acres (17 percent) of wetlands were determined “degraded” by the applicant.

Wetland types are further categorized in the table below:

Table 4: Wetland Types for Route A

Wetland Types for Route A		
Wetland Type	**Individual Size (acres)**	**Total Size (acres)**
Degraded Floodplain Forest	0.3	1.5
Floodplain Forest	1.2	
Degraded Fresh (Wet) Meadow	3.0	13.5
Fresh (Wet) Meadow	10.5	
Degraded Hardwood Swamp	0.4	3.0
Hardwood Swamp	2.6	
Degraded Seasonally Flooded Basin	0.2	12.9
Seasonally Flooded Basin	12.7	
Degraded Sedge Meadow	0.2	0.2
Degraded Shallow Marsh	1.7	2.9
Shallow Marsh	1.2	
Degraded Shrub-Carr	0.0	1.5
Shrub-Carr	1.5	
Total	**35.5 acres**	

Generally speaking,

- The forested wetlands may be found as isolated features surrounded by agricultural land; part of a larger, higher quality wetland complex that is crossed by an existing overhead transmission line corridor; or located within riparian corridors near higher-quality waterways. The forested wetlands range from having a dominance and diversity of native species to a having a mixture of native and invasive species, such as reed canary grass. Overall, the forested wetlands may be considered to have medium to high functional values, as they may provide stormwater and flood storage adjacent to waterways and agricultural fields; support wildlife foraging/habitat; support water quality, shoreline, and flood protection within riparian corridors; support groundwater recharge/discharge; and provide aesthetic and recreational value within riparian corridors.

- The shrub-scrub wetlands may be found surrounded by agricultural land; along road ROW; or part of a larger, higher-quality wetland complex that is crossed by an existing overhead transmission line corridor. The shrub-scrub wetlands are generally dominated by invasive species, such as buckthorn and reed canary grass; some of the scrub-shrub wetlands within greater wetland complexes have a greater diversity of plant species, native and invasive. Overall, the shrub-scrub wetlands may be considered to have low to medium functional values, as they may provide stormwater and flood storage adjacent to road ROW and agricultural fields; support wildlife foraging/habitat; and support water quality, shoreline, and flood protection within riparian corridors.

- The emergent wetlands may be found within or immediately adjacent to active agricultural land and along road ROW. The emergent wetlands tend to be dominated by invasive plant species (such as reed canary grass or hybrid cattails) or by agricultural crops and have little floral diversity. Some of the emergent wetlands that are part of a larger wetland complex that are not disturbed by active farming practices may have slightly greater presence of native plants and greater floral diversity. Overall, the emergent wetlands may be considered to have low to medium functional values, as they may provide stormwater and flood storage adjacent to road ROW and agricultural fields; support wildlife foraging/habitat; support groundwater recharge/discharge; and support water quality, shoreline, and flood protection within riparian corridors.

Wetlands designated as areas of special natural resource interest (ASNRI) may be present within or adjacent to Route A, including wetlands that may provide suitable habitat for state special concern, threatened, and endangered species; wetlands adjacent to White River State Trail, a DNR-owned/managed property; and wetlands that are located adjacent to or hydrologically connected to the Root River and its tributaries. The Root River and its tributaries crossed by Route A are designated as ASNRI "Coastal Wisconsin Wetland Waters" due to the presence of high-quality wetlands and riverine forest along and adjacent to the waterway.

Route B

Approximately 185 wetlands, totaling 35.6 acres, were identified for Route B. Wetland types were identified as 0.5 acres of Floodplain Forest, 16.8 acres of Fresh (Wet Meadow), 3.0 acres of Hardwood Swamp, 9.2 acres of Seasonally Flooded Basin, 4.7 acres of Shallow Marsh, 1.3 acres of Shrub-Carr, and 0.1 acres of Wet Mesic Prairie. Wetland community types were identified as 3.5 acres of forested wetland, 1.3 acres scrub-shrub wetland, and 30.8 acres emergent wetland.

The applicant further categorized the wetland types as either degraded or not degraded, as shown in the table below. Approximately 11.9 acres (33 percent) of wetlands were determined "degraded" by the applicant.

Table 5: Wetland Types for Route B

Wetland Type	Individual Size (acres)	Total Size (acres)
Degraded Floodplain Forest	0.5	0.5
Degraded Fresh (Wet) Meadow	6.0	16.9
Fresh (Wet) Meadow	10.9	
Degraded Hardwood Swamp	0.8	3.0
Hardwood Swamp	2.2	
Degraded Seasonally Flooded Basin	0.6	9.2
Seasonally Flooded Basin	8.6	
Degraded Shallow Marsh	3.5	4.7
Shallow Marsh	1.2	
Degraded Shrub-Carr	0.6	1.3
Shrub-Carr	0.7	
Wet Mesic Prairie	0.1	0.1
Total	**35.6 acres**	

Generally speaking,

- The forested wetlands are mostly found surrounded by agricultural land as part of larger wetland complex; within higher-quality riparian corridors; or part of a larger, higher quality wetland complex. Within and surrounded by agricultural lands, the forested wetlands tend to be dominated by invasive species below the canopy and part of a greater wetland complex where other wetland community types are dominated by invasive species or lack floristic diversity due to agricultural crops; the more isolated forested wetlands or those hydrologically connected to waterways or higher-quality wetland complexes tend to have a greater floristic diversity and greater presence of native plant species. Overall, the forested wetlands may be considered to have medium to high functional values, as they may provide stormwater and flood storage adjacent to waterways and agricultural fields; support wildlife foraging/habitat; support water quality, shoreline, and flood protection within riparian corridors; support groundwater recharge/discharge; and provide aesthetic and recreational value within riparian corridors.

- The shrub-scrub wetlands may be found as part of a larger, wetland complex with multiple wetland community types, adjacent to agricultural lands; immediately adjacent or within road ROW; or part of a larger, higher-quality wetland complex serving as a riparian corridor and crossed by an existing overhead transmission line corridor. Within more disturbed areas, the shrub-scrub wetlands may be dominated by herbaceous invasive species, such as hybrid cattail and reed canary grass; within riparian corridors or within larger wetland complexes, there tends to be a greater diversity of plant species, native and invasive. Overall, the shrub-scrub wetlands may be considered to have low to

medium functional values, as they may provide stormwater and flood storage; support wildlife foraging/habitat; support water quality; and support shoreline and flood protection within riparian corridors; and provide aesthetic and recreational value within riparian corridors.

- The emergent wetlands may be found within or immediately adjacent to active agricultural land and along road ROW; some emergent wetlands are isolated features or may be part of a larger wetland complex. The emergent wetlands tend to be dominated by invasive plant species (such as reed canary grass) or by agricultural crops and have little floral diversity. Overall, the emergent wetlands may be considered to have low to medium functional values, as they may provide stormwater and flood storage adjacent to road ROW and agricultural fields, groundwater recharge, and support wildlife foraging/habitat.

Wetlands designated as ASNRI may be present within or adjacent to Route B, including wetlands that may provide suitable habitat for state special concern, threatened, and endangered species; wetlands that are located adjacent to or hydrologically connected to the Root River and its tributaries; and wetlands adjacent to or hydrologically connected to the DNR-owned/managed properties White River State Trail and Richard Bong Recreation Area.

Route C

Approximately 40 wetlands, totaling 10.5 acres, were identified for Route C, which, unlike proposed Routes A and B, is comprised of several individual segments that cannot be combined to create a single contiguous route option. Wetland types were identified as 5.0 acres of Fresh (Wet Meadow), 0.7 acres of Hardwood Swamp, 4.3 acres of Seasonally Flooded Basin, 0.3 acres of Shallow Marsh, 0.1 acres of Shrub-Carr. Wetland community types were identified as 0.7 acres of forested wetland, 0.1 acres scrub-shrub wetland, and 9.6 acres emergent wetland.

The applicant further categorized the wetland types as either degraded or not degraded, as shown in the table below. Approximately 1.1 acres (10 percent) of wetlands were determined "degraded" by the applicant.

Table 6: Wetland Type for Route C

Wetland Type	Individual Size (acres)	Total Size (acres)
Degraded Fresh (Wet) Meadow	0.9	5.0
Fresh (Wet) Meadow	4.1	
Degraded Hardwood Swamp	0.3	0.8
Hardwood Swamp	0.5	
Seasonally Flooded Basin	4.3	4.3
Shallow Marsh	0.3	0.3
Shrub-Carr	0.1	0.1
Total	**10.5 acres**	

Generally speaking,

- The forested wetlands are mostly found surrounded by agricultural land as part of larger wetland complex. The forested wetlands tend to be dominated by invasive species below the canopy and part of a greater wetland complex where other wetland community types are dominated by invasive species or lack floristic diversity due to agricultural crops. Overall, the forested wetlands may be considered to have medium functional values, as they may provide stormwater and flood storage within and adjacent to road ROW and agricultural fields; support wildlife foraging/habitat; and support groundwater recharge.

- The shrub-scrub wetland is part of a larger, emergent (wet marsh, fresh wet meadow) wetland complex surrounded by agricultural lands. Vegetation present within the wetland complex was not documented. The shrub-scrub wetland may be considered to have medium functional values, as the wetland complex may provide stormwater and flood storage, support wildlife foraging/habitat, and support groundwater recharge.

- The emergent wetlands may be found within or immediately adjacent to active agricultural land and along road ROW; the wetlands may be isolated features or part of a larger wetland complex. The emergent wetlands tend to be dominated by agricultural crops or anticipated to be dominated by invasive plant species (such as reed canary grass) and have little floral diversity (limited vegetation data is available). Overall, the emergent wetlands may be considered to have low to medium functional values, as they may provide stormwater and flood storage adjacent to road ROW and agricultural fields, support groundwater recharge, and support wildlife foraging/habitat.

Wetlands designated as ASNRI may be present within or adjacent to Route C, including wetlands that may provide suitable habitat for state special concern, threatened, and endangered species; and wetlands adjacent to or hydrologically connected to the DNR-owned/managed Richard Bong Recreation Area.

Staging Areas

Approximately 19 wetlands, totaling 2.1 acres, were identified within proposed staging areas. Wetland types were identified as 1.5 acres of Fresh (Wet Meadow), 0.02 acres of Hardwood Swamp, 0.1 acres of Seasonally Flooded Basin, 0.4 acres of Shallow Marsh, 0.03 acres of Shrub-Carr. Wetland community types were identified as 0.02 acres of forested wetland, 0.03 acres scrub-shrub wetland, and 2.0 acres emergent wetland.

The applicant further categorized the wetland types as either degraded or not degraded, as shown in the table below. Approximately 1.0 acre (48 percent) of wetlands was determined “degraded” by the applicant.

Table 7: Wetland Type for Staging Areas

Wetland Type	Individual Size (acres)	Total Size (acres)
Degraded Fresh (Wet) Meadow	0.7	1.5
Fresh (Wet) Meadow	0.8	
Hardwood Swamp	0.03	0.02
Seasonally Flooded Basin	0.1	0.1
Degraded Shallow Marsh	0.4	0.4
Shrub-Carr	0.03	0.03
Total	**2.1 acres**	

Generally speaking,

- The wetlands are located adjacent to road ROW, within active agricultural land, and/or within or adjacent to previously disturbed, impervious areas (such as existing parking areas, storage areas, etc.).

- The wetlands tend to be dominated by agricultural crops or invasive species. Overall, the wetlands may be considered to have low functional values, as they may provide stormwater and flood storage within and adjacent to road ROW and agricultural fields and support wildlife foraging/habitat.

Wetlands designated as ASNRI are not expected to be present within or adjacent to staging areas unless the wetlands would provide suitable habitat for state special concern, threatened, and endangered species.

5.4.2 Potential Wetland Impacts

Construction activities conducted near and across wetlands have the potential to impact wetland functional values, such as floristic diversity; fish and wildlife habitat; flood protection; water quality protection; shoreline protection; groundwater processes; and aesthetics, recreation, education, and science.

For example, disturbances in and adjacent to wetlands (e.g. vegetation clearing, siting of project components, areas of land disturbance, etc.) could lead to:

- Increases of invasive species and decreases in native species diversity.

- Impacts to wildlife habitat and corridors from vegetation clearing, fragmentation, and the siting of project components.

- Impacts to fish and aquatic invertebrates from the clearing of riparian wetlands that may provide shade, food, and filter pollutants from entering waterways.

- Reduction of the capacity for wetlands to capture, detain, and/or filter stormwater and run-off from adjacent agricultural land.

- Reduction of the natural water quality benefit of wetlands.
- Reduction of the natural buffer, stabilization, and erosion control along waterways.
- Reduction of aesthetics and recreational opportunities.

The degree and nature of impacts to wetlands depends on multiple factors, such as the characteristics of the wetlands, quality of the wetlands, ground conditions at the time of construction, and the type and duration of construction activities. Short-term wetland impacts can become long-term impacts if the construction phase is not well managed or if restoration techniques are not properly applied.

The applicant is proposing wetland impacts from excavation activities (e.g. open cut trenching, bore pits), placement of construction matting, and forested wetland conversion. The applicant is proposing to avoid regulated wetland impacts within staging areas.

5.4.2.1 Temporary Wetland Fill

Wetland soils consist of primarily organic matter (decomposed plant material) which forms very slowly. If disturbed by digging, filling, and/or compaction, these soils may not readily recover and may not easily be repaired. Compaction from heavy equipment or from placement of construction matting in wetland can impact surface and/or groundwater flow. Soil compaction reduces the water-holding capacity of the soil and may result in increased runoff. Compacted soils could result in a change in vegetation by reducing plant diversity and promoting the growth of invasive species. Operating equipment in wetland can also endanger amphibians and other aquatic life.

Soil compaction could be minimized through the use of construction matting and constructing during frozen or dry, stable ground conditions. Construction matting is utilized to spread the distribution of equipment weight when crossing wetlands during the growing season or when wetlands are not stable or not frozen, as heavy machinery used for construction could crush wetland vegetation and damage wetland soils, causing soil compaction, rutting, and/or soil mixing.

5.4.2.2 Placement of Construction Matting

The applicant is proposing to place temporary construction matting in wetlands in order to facilitate vehicle access and operation. Approximately 28.7 acres of construction matting is proposed in wetlands for Route A, approximately 27.6 acres of construction matting is proposed in wetlands for Route B, and approximately 7.9 acres of construction matting is proposed in wetlands for Route C.

The applicant stated temporary construction matting is not anticipated to be placed in wetlands for greater than 60 consecutive days between May 15th and November 15th. If project plans change, and matting would be placed in matting for greater than 60 consecutive days during the

growing season, the applicant would be required to submit a wetland restoration plan for these areas, subject to DNR review and approval.

5.4.2.3 Excavation Activities

The applicant is proposing to impact wetlands via excavation in order to install the natural gas pipeline; installation methods include directional boring or open-cut trenching within wetlands. Approximately 2.7 acres of wetland excavation is proposed for Route A, approximately 2.7 acres is proposed for Route B, and approximately 0.7 acres is proposed for Route C.

Temporary wetland fill impacts are presented in the table below:

Table 8: Temporary Wetland Fill

Temporary Wetland Fill			
Route	**Fill from Construction Matting (acres)**	**Fill from Excavation Activities (acres)**	**Total Temporary Wetland Fill (acres)**
A	28.7	2.7	**31.4**
B	27.6	2.7	**30.4**
C	7.9	0.7	**8.6**

5.4.2.4 Stone Ditch Checks

The applicant is proposing to install temporary stone ditch checks within wetland W-563 due to the steep slopes and high rate of stormwater flow expected at this location; the stone ditch check is expected to help control the flow and protect sensitive resources downslope and ultimately result in less impact to the wetland compared to other less effective measures.

The applicant states the stone ditch checks would be installed by placing geotextile fabric by hand and using a backhoe or excavator (located outside of the wetland, if possible) to place stone on top of the fabric. Once construction at this location is completed, the stone and fabric would be removed and disposed of in accordance with applicable permits or regulations.

Wetland impacts would be minimized by staging machinery outside of the wetland and utilizing timber matting within the wetlands if any machinery is required to be located closer to the ditch check locations. Additionally, the geotextile fabric would be installed by hand and any stone left after the removal of the fabric (with the stone inside) would also be removed by hand.

5.4.2.5 Permanent Wetland Fill

Permanent wetland fill results in the loss of wetland acreage and the functional values the wetlands provided. The applicant states the project would not result in permanent wetland fill.

5.4.2.6 Wetland Conversion

Clearing of wetlands dominated by woody vegetation results in a conversion from shrub-scrub and forested wetland into emergent wetland and may impair the wetlands' functional values, as described above.

Clearing and removal of brush and trees is proposed by the applicant within temporary workspaces and permanent corridor ROW to facilitate equipment access and construction of the pipeline and ensure safe clearances between vegetation and the pipeline. In shrub-scrub and forested wetlands proposed to be crossed via open-cut trenching, permanent conversion to emergent wetland would occur within a 20-foot-wide path over the pipeline. In shrub-scrub and forested wetlands proposed to be crossed via boring, the applicant states permanent forested wetland conversion would be avoided and woody vegetation clearing would be limited to a 10-foot path along the bore path to facilitate guiding the drill head during construction and future surveys. The 10-foot path of woody vegetation clearing would be limited to smaller woody vegetation that would be hand-cut or trimmed, leaving larger-diameter canopy trees in place.

In areas of permanent conversion, these areas would be maintained in herbaceous wetland state in perpetuity to accommodate inspections and surveys of the pipeline corridor. Temporary workspaces would be cleared of shrubby and woody vegetation, as needed, and would be allowed to revegetate once construction is completed. The applicant is not proposing to replant woody vegetation within wetlands once cleared.

The applicant states wetland clearing activities would take place during dry or frozen ground conditions, when possible, using low ground pressure tires or tracks, and/or utilizing construction matting to prevent soil disturbance.

Vegetation would be cut at or slightly above the ground surface using mechanized mowers, processers, harvesters, or by hand. The applicant states rootstocks would generally be left in place except in areas where stump grinding is necessary to facilitate the movement of construction vehicles or as requested by landowners. If rootstocks would be removed or stump griding would occur within wetlands, this would be a regulated wetland impact that would require DNR approval prior to conducting this work. At this time, the applicant has not included rootstock removal or stump griding in their DNR Table 1, Wetland and Waterway Impact/Crossing Table.

The applicant proposes to remove larger brush and woody debris from wetlands and thinly and uniformly scatter smaller fragments and chipped debris within wetlands in a manner that would not impede vegetation growth, water flow, or elevations within the wetland.

Approximately 4.1 acres of wetland conversion (3.8 acres temporary conversion, 0.3 acres permanent conversion) is proposed for Route A, approximately 3.3 acres of wetland conversion (2.3 acres temporary conversion, 1.0 acres permanent conversion) is proposed for Route B, and approximately 0.3 acres wetland conversion (0.2 acres temporary conversion, 0.1 acres permanent conversion) is proposed for Route C.

Wetland conversion by wetland classification type is displayed in the table below:

Table 9: Wetland Conversion

Wetland Conversion			
Route	**Shrub-Scrub Wetland**	**Forested Wetland**	**Total (acres)**
A	0.7	3.4	4.1
B	0.9	2.3	3.2
C	0.0	0.3	0.3

5.4.3 Wetland Impact Avoidance and Minimization

5.4.3.1 Wetland Avoidance

All attempts should first be made to avoid impacting wetlands. Impacts to wetlands could be avoided by routing the natural gas pipeline, crossing wetlands completely via boring, avoiding equipment access across wetlands, and siting access roads and staging areas outside of wetlands.

The applicant stated all proposed route segments have been selected to avoid and minimize wetland impacts to the extent practicable, however, given the extent of wetlands in the project area, proximity to residential homes and farming/livestock operations, and restrictions on how close land disturbance can take place near existing overhead transmission structures, wetland impacts cannot be completely avoided.

The applicant stated they evaluated and implemented wetland avoidance measures during initial route planning stages, for example re-routing specific areas of the route to avoid forested wetland areas or shifting specific areas of the route to avoid setting up boring equipment in non-farmed wetlands. The applicant also provided some site-specific examples of how they evaluated project components in relation to wetlands and provided either discussion on how project component siting was updated to avoid or minimize impacts or provided discussion on why impacts to the wetland could not be avoided.

The applicant stated open-cut trenching is a preferred pipeline installation method compared to boring due to the risk of inadvertent releases (IRs), an increase in risk of IRs when installing larger-diameter pipelines (this can result in greater drilling distances or depths to cross a resource), creating access challenges for pipeline maintenance digs (the pipeline could be buried deeper in the ground during boring installation), and differences in costs (boring is more expensive). The applicant also states the majority of open-cut trenching within wetlands would occur in lower-quality, farmed wetlands that provide lower functional values.

5.4.3.2 Wetland Minimization

Where complete wetland avoidance is not possible due to engineering constraints, existing infrastructure, or other factors, wetland impacts should be minimized as much as possible.

Construction methods that can minimize impacts to wetlands include:

- Conducting construction activities when wetland soils were frozen or stable and vegetation is dormant.
- Using construction matting and wide-track vehicles with equipment crossing of wetlands when wetlands were not stable or not frozen.
- Using adjacent roadways and existing off-ROW access roads for access, when possible.
- Siting structures and access roads on the edges of wetlands rather than in the middle of wetland to avoid fragmenting wetland complexes.
- Reducing the construction workspace in wetlands.
- Installing site-specific sediment and erosion control measures and devices prior to construction activities, with daily inspections and maintenance throughout all construction and restoration phases.
- Implementing a construction sequencing plan that minimizes the amount of land disturbed or exposed (susceptible to erosion) at one given time across the project.
- Marking the boundary of wetlands.
- Preparing and implementing an invasive species management plan that identifies known areas of invasive species populations, addresses site restoration activities, and includes specific protocols to minimize the spread of invasive species.
- Minimizing the amount of vegetation clearing in wetland and conversion of wetland types.
- Limiting forested wetland clearing areas to the permanent ROW corridor.
- Removing all brush piles, wood chips, and woody debris from wetlands following clearing activities.
- Removing woody debris (including wood chips) from wetlands located in FEMA-mapped floodplain.
- Conducting surface and sub-surface assessments prior to construction, including hydrology and soil evaluations. This includes modifying the engineering plans, as needed, to avoid and minimize long term impacts to surface and subsurface resources and to re-establish conditions post-construction.
- Preparing and implementing dewatering practices that prevent sedimentation into wetlands.

- Revegetating disturbed areas and areas of exposed soil as soon as possible and seeding with a cover crop and/or native seed mix to help prevent the establishment of invasive species.

- Scheduling construction to avoid disrupting sensitive species.

- Limiting the amount of time necessary to complete construction.

The applicant has or is proposing to reduce impacts to wetlands within the project area by implementing the following:

- Avoiding wetland impacts in staging areas and areas of equipment storage.

- Avoiding topsoil stripping within access roads that cross/occur in wetlands.

- Avoiding topsoil stripping outside of the trench line when working within wetlands.

- Completing open-cut trenching in wetlands in a timely manner.

- Stockpiling excavated soils on matting when working in wetlands.

- Segregating topsoil and subsoil within wetlands during excavation.

- Locating dewatering structures in uplands to the greatest extent practicable.

- Crossing wetlands via boring, with bore pits located outside of wetlands, where possible. The applicant is proposing to cross ASNRI-designated wetlands along Route A via boring.

- Using specialty native seed mixes within disturbances of high-quality habitats, including wetlands and riparian zones.

- Avoiding the use of rock access pads in wetlands.

- Conducting additional investigations/evaluations to better plan for horizontal directional drilling (HDD) installation across/adjacent to high-quality wetlands and waterways.

- Avoiding permanent forested wetland conversion within areas of boring.

- Utilizing construction matting in all wetlands outside of any trench or other pit excavations.

- Reducing the temporary workspace width near specific, high-quality wetlands, where possible.

- Shifting areas of the route to cross narrower portions of wetland.

- Proposing to shift bore pits outside of wetlands or extend bore paths (upon final design evaluation and approvals) for higher quality wetlands W-084, W-071, W-260.

5.4.4 Wetland Restoration

Site restoration consists of returning the areas impacted by the construction back to their original condition, if not better. Restoration typically occurs in any disturbed areas within easements or ROW, temporary construction areas, staging areas, off-ROW access roads, and any other areas used for Project related activities, with the exception of areas of permanent wetland impact. Site restoration, including re-vegetation, of the disturbed areas should be completed as soon as possible following construction. Sediment and erosion control devices should be installed before ground disturbance occurs to reduce erosion and trap sediment from entering sensitive resources and should be in place until vegetation is re-established.

The applicant stated that following construction, all temporarily impacted wetlands would be restored to pre-existing conditions through re-vegetation and restoration plans. Restoration would include restoring wetland grades to original elevations, allowing trees and shrubs to revegetate in areas of temporary wetland conversion, and utilizing specialty native seed mixes within high-quality habitats that were disturbed during construction (including wetlands and riparian zones).

Farmed wetlands would not be re-seeded due to their current land use.

The applicant states the following restoration techniques would be implemented after construction is completed:

- Removal of all construction-related materials (e.g. timber matting) and the restoration of significant ruts and depressions.

- Soil removed and separated by profile during open-cut trenching would be restored to the same profile following pipe installation.

- The ROW would be restored to pre-existing topography as much as practicable. Areas with significant rutting in wetlands would be repaired using hand tools, back dragging, or other appropriate means to restore topography while minimizing additional disturbance.

- Erosion controls would be maintained, and bare soil stabilized to be compliant with Wis. Admin. Code chs. NR 216 and 151 technical standards. A temporary cover crop may be installed over disturbed soils following ground disturbance.

- If conditions are adequate, wetlands would be allowed to revegetate with existing seed bank before alternative seeding is used. Under certain circumstances, a cover crop or a native wetland seed mix may be necessary.

- The vegetation composition objective is to match pre-existing dominant species with no new invasive species.

- Wetlands would be monitored to document germination success and community composition. Sites would be revisited regularly to note vegetative cover response until objectives are met.

- If adequate, uniform cover is not achieved within a growing season then the site would be reassessed to determine if reseeding is needed.

5.4.4.1 Wetland Restoration, Post-Construction Monitoring Plans

The DNR would require the applicant to develop a detailed post-construction wetland and waterway restoration and monitoring plan and a detailed invasive species management and monitoring plan for DNR review and approval prior to beginning construction.

5.5 Waterways

Waterways in the form of creeks, streams, rivers, and lakes are abundant throughout Wisconsin, and provide for many recreational activities, as well as habitat for aquatic species. Wisconsin has more than 12,600 rivers and streams that meander their way through 84,000 miles of varying terrain. About 32,000 miles of these streams perennially or continuously run throughout the year, while the remainder flow intermittently during spring and other high-water times.

The proposed project is located within the Headwaters Des Plaines River watershed (HUC 0712000401), Eagle Creek-Fox River watershed (HUC 0712000607), Oak Creek-Frontal Lake Michigan watershed (0404000201), Wind Lake Drainage Canal watershed (HUC 0712000603), Root River Canal watershed (HUC 0404000202), Manitou Creek-Fox River watershed (HUC 0712000610), and Root River watershed (HUC 0404000203).

5.5.1 Waterway Identification and Quality

Waterways were identified by evaluating aerial imagery, the 24K hydro layer of the DNR Surface Water Data Viewer, and field observations during the wetland field surveys.

Route A: Approximately 33 navigable waterways were identified for Route A. Seven of the waterways are designated as PNW-ASNRI Coastal Wisconsin Wetlands Streams (The Root River and its tributaries crossed by Route A are designated as ASNRI "Coastal Wisconsin Wetland Waters" due to the presence of high-quality wetlands and riverine forest along and adjacent to the waterway). No waterways are designated as trout streams.

Route B: Approximately 31 navigable waterways and three waterbodies were identified for Route B. None of the waterways or waterbodies are designated as ASNRI or trout streams. One waterway, an unnamed tributary (WBIC: 5038874) Unnamed Tributary was determined non-navigable by DNR.

Route C: Approximately seven navigable waterways were identified for Route C. None of the waterways or waterbodies are designated as ASNRI or trout streams.

Staging Areas: No navigable waterways are located within staging areas.

5.5.2 Potential Waterway Impacts

Construction activities conducted near and across waterways would have the potential to impact public interests in navigable waterways, such as water quality, aquatic species habitat, recreation, navigation, and scenic beauty of the resource. Forested and shrub areas along waterways provide a natural corridor for wildlife movement, help maintain soil moisture levels in waterway banks, provide bank stabilization, filter nutrient-laden sediments and other runoff, maintain cooler water temperatures, and encourage a diversity of vegetation and wildlife habitats. The removal of riparian vegetation may cause water temperatures to rise and negatively affect aquatic habitats, especially cold-water systems. Removing riparian wetland vegetation may also decrease shoreline protection and could lead to increased sedimentation to waterways. Vegetation disturbance along the waterway may also lead to the infestation by invasive and nuisance species. Aesthetics within and along waterway and riparian corridors could be affected by new transmission facilities if where vegetation clearing is proposed.

The use of heavy equipment on waterway banks could cause soil compaction. Constructing in areas with seeps and springs could temporarily alter the surface and subsurface hydrology feeding waterways.

The degree and nature of impacts to waterways depends on multiple factors, such as the characteristics of the waterways, quality of the waterways, ground conditions at the time of construction, and the type and duration of construction activities. Short-term and waterway impacts can become long-term impacts if the construction phase is not well managed or if restoration techniques are not properly applied.

The applicant is proposing to install the pipeline across waterways via boring or open-cut trenching. The applicant is also proposing to install temporary clear span bridges (TCSB's) across waterways for equipment access.

In order to protect fish spawning habitat, TCSBs and any in-water work activities would not be authorized during the fish spawning timing restriction periods, unless the local DNR Fisheries Biologist determines fish-timing restrictions are waived.

5.5.2.1 Waterways and Waterbodies Greater than 35 feet wide OHWM to OHWM, Temporary Bridges, and Open-Cut Trenching

Apart from one waterway and one waterbody along Route B, all waterways proposed for a TCSB crossing or pipeline installation via open-cut trenching would be less than 35 feet wide Ordinary High-Water Mark to Ordinary High-Water Mark (OHWM).

Unnamed Tributary to Eagle Lake (identified as Waterway S-50 by the applicant) is a perennial waterway estimated to be approximately 46 feet wide OHWM to OHWM. An unnamed

waterbody (identified as Waterbody WB-20 by the applicant) is estimated to be approximately 201 feet wide OHWM to OHWM. Both surface waters are proposed to be crossed via temporary bridges and open-cut trenching for pipeline installation.

The applicant stated waterway S-50 and waterbody WB-20 could be crossed by HDD, however, many factors were considered when determining the construction methods: when taking into account site specific characteristics including the quality of the resource (agricultural field ditch and agricultural pond), as well as general considerations discussed in Section 5.5.3 Waterway Impact Avoidance and Minimization, open-cut trenching was determined to be the best crossing method for pipeline installation. The applicant also stated open-cut trenching across S-50 and WB-20 would be dependent on-site conditions at the time of construction and the open-cut trenching is proposed during optimal weather conditions and/or times of year to take advantage of lower water levels. If site conditions are not favorable for open-cut trenching based on water levels, that applicant would consider alternative crossing methods, including boring.

For these waterway and waterbody crossings, the DNR considered the following during their review of the CA application materials

- How pipeline installation could be completed logistically, especially for the 200-foot-wide waterbody.

- How the open-cut trenching could be completed in a manner that would avoid short-term and long-term adverse impacts to water quality, fisheries, flow, and other public interests.

- How the open-cut trenching could be completed in a safe manner, especially during active construction if there are unforeseen weather events and high incoming flows.

- How the open-cut trenching could be completed and the bed and banks restored in a manner that would not result in long-term impacts to the resource.

- How temporary bridges could be installed and utilized logistically, especially for the 200-foot-wide waterbody.

- How temporary bridges could be installed safely and support the equipment and vehicle crossings over a larger distance.

- What alternatives exist for equipment and vehicles to access either side of the waterway and waterbody without the need for a temporary bridge.

- Whether an engineered bridge would provide greater support for vehicles and equipment rather than a traditional temporary bridge.

Detailed plans for crossing waterway S-50 and WB-20 via temporary bridges and open-cut trenching have not yet been developed by the applicant, but preliminary plans were submitted as part of the Plan & Profile Maps (PSC Reference 525526).

At this time, the applicant has not demonstrated to DNR that the open-cut trenching and temporary bridging of these surface waters would be the least impactful construction methods on the resources. If the project is approved and a route selected, the applicant has committed to completing additional field work at these locations and developing final, detailed crossing plans. These plans and designs would need to be reviewed and approved by DNR before beginning construction.

If open-cut trenching or placement of temporary bridges is proposed for these surface waters, the applicant should provide, at a minimum, the following information as part of the detailed crossing plans:

- The site conditions, including elevations and substrate information, for the surface waters that were collected during field surveys.
- Details on flow bypass and work zone isolation systems, such as type of bypass/isolation systems, locations of structures, sizes of structures, pumping rates, locations of dewatering structures, etc.
- Anticipated water levels during construction.
- Location(s), sizes, and designs for temporary bridges, including any in-stream support structures.
- Contingency plans in the case water levels rise, incoming flow increases, unforeseen weather events, etc.
- Detailed restoration designs for the surface waters post construction.

5.5.2.2 Temporary Clear Span Bridges

The proposed TCSB's would consist of construction mats and/or steel I-beam frames, or other similar material, approximately 16 feet wide or longer. The TCSBs would be placed above the OHWM on either side of the waterway banks to completely span the waterways. The TCSB's would not require a support structure on the bed of the waterways. The applicant proposes minor blading or excavation above the OHWM in order to safely install the bridges on the banks; the removal of low-growing trees, shrubs, and other shoreline vegetation would be kept to a minimum; proper erosion control measures (e.g. straw bales/logs, silt fencing, etc.) would be implemented and maintained during and after the use of the TCSB to prevent runoff or siltation into the waterway; the TCSB's would be secured to a fixed anchor and inspected routinely while it is installed; the TCSBs would be visually inspected and have any debris or mud removed daily; and monitoring reports detailing the inspections of waterway/TCSB crossings would be maintained throughout the life of the project.

The applicant states TCSBs would be installed for as long as access is needed during construction and would be removed and the area restored once construction has been completed in the area and access across the waterway is no longer required; depending upon the level of disturbance, restoration may include minor grading/leveling to restore pre-existing topography, installation of seed, and stabilizing the banks with erosion control measures such as erosion mats and straw logs.

The impacts of the TCSB placements and removals would be minimal if constructed properly. Potential impacts would be expected to be short-term and may include disturbance to the banks of the waterways; clearing of riparian vegetation; disruption to the invertebrates, fish, and/or wildlife associated with the waterways during installation, use, and removal of the TCSBs; sedimentation into the waterway; and/or public access limitations. If improperly installed or maintained, TCSB's could damage banks and cause erosion or be overtopped or dislodged and back up water. TCSB's have the potential to temporarily impact navigation in the waterway; under NR 320.04(1), Wisconsin Administrative Code, a five-foot clearance shall be maintained between the water and TCSB, unless the requirements in NR 320.04(3) would be met, including providing portage for anyone navigating the waterway.

Approximately 19 TCSBs would be installed across navigable waterways for Route A, 19 TCSBs for Route B (excluding two temporary bridges for WB-20 and S-50), and five TCSBs for Route C. The remaining waterways would not be traversed with equipment.

5.5.2.3 Open-Cut Trenching (Dredging)

The applicant is proposing to install the natural gas pipeline across waterways via boring and open-cut trenching. Open-cut trenching within a waterway, results in removal or disturbance of material from the bed and is therefore regulated by DNR as "dredging" under Wis. Stat. ch. 30.20.

Temporary structures, in the form of flow bypass/work isolation systems, would be installed on the bed of the waterway prior to open-cut trenching the waterway to isolate the area of disturbance. Unless the waterway to be crossed would be completely dry below the OHWM for the entire duration of the in-water work, including accounting for rain events during construction, the DNR would require open-trenching in the waterway to be completed using a work zone isolation/flow bypass approach appropriate for the conditions of the waterway.

Wisconsin Admin. Code § NR 347.05 requires that anyone seeking to remove materials from the beds of waterways shall provide the DNR with preliminary information, including:

a) Name of waterbody and location of project;

b) Volume of material to be dredged;

c) Brief description of dredging method and equipment;

d) Brief description of proposed disposal method and location and, if a disposal facility is to be used, size of the disposal facility;

e) Any previous sediment sampling (including field observations) and analysis data from the area to be dredged or from the proposed disposal site;

f) Copy of a map showing the area to be dredged, the depth of cut, the specific location of the proposed sediment sampling sites and the bathymetry of the area to be dredged; and

g) Anticipated starting and completion dates of the proposed project.

The DNR received preliminary information from the applicant in April 2025 and will conduct an initial evaluation to determine if there is reason to believe that the material proposed to be dredged is contaminated and whether further sediment sampling and analysis is required.

Approximately 16 navigable waterways are proposed for open-cut trenching pipeline installation for Route A, 19 navigable waterways and one waterbody for Route B, and four navigable waterways for route C.

5.5.3 Waterway Impact Avoidance and Minimization

5.5.3.1 Waterway Avoidance

All attempts should first be made to avoid impacting waterways. Impacts to waterways could be avoided by routing the natural gas pipeline away from riparian corridors, crossing all waterways via boring, and utilizing alternate access routes to access project components.

The applicant stated open-cut trenching is a preferred pipeline installation method compared to boring due to the risk of IRs, an increase in risk of IRs when installing larger-diameter pipelines (this can result in greater drilling distances or depths to cross a resource), creating access challenges for pipeline maintenance digs (the pipeline could be buried deeper in the ground during boring installation), and differences in costs (boring is more expensive). The applicant also states that during preliminary internal routing meetings, routes and segments were compared, taking into account various criteria, including environmental impacts.

The applicant also states most waterways that would be open-cut trenched for pipeline installation were identified as low-quality, constructed and farmed drainage features and that boring installation across waterways is primarily proposed at locations to minimize resource impacts to moderate to high-quality waterways.

As currently proposed, the portion of the Root River and its tributaries designated as PNW/ASNRI would be crossed via boring; boring completely underneath waterways is not a DNR-regulated waterway activity and should not result in regulated waterway impacts if boring is successful.

5.5.3.2 Waterway Minimization

Where complete waterway avoidance would not be possible due to engineering constraints, existing infrastructure, or other factors, waterway impacts should be minimized as much as possible. Construction activities and pipeline installation across waterways may have both short-term and long-term impacts. The type and significance of the impact is dependent on the characteristics of the waterway, how construction is completed, and how the resources are stabilized and restored.

Construction methods that can minimize impacts to waterways include:

- Installing the pipeline via boring across surface waters S-50 and WB-20 and avoiding the installation of TCSBs across surface waters S-50 and WB-20, which are greater than 35 feet wide OHWM to OHWM.
- Marking waterway boundaries.
- Minimizing the number of potential vehicle crossings of waterways by accessing the ROW on either side of the stream or from adjacent roads.
- Installing site-specific sediment and erosion control measures and devices prior to construction activities, including daily inspections and maintenance throughout all construction and restoration phases. This includes installing sediment control BMP's under and on the sides of TCSB's.
- Implementing a construction sequencing plan that minimizes the amount of land disturbed or exposed (susceptible to erosion) at one given time across the project.
- Leaving existing vegetative buffers undisturbed whenever possible or minimizing vegetation clearing in riparian zones.
- Revegetating disturbed areas and areas of exposed soil as soon as possible.
- Maintaining shaded stream cover.
- Avoiding the use of herbicides near waterways or utilizing herbicides approved for use in aquatic environments.
- Conducting surface and sub-surface assessments prior to construction, including hydrology and soil evaluations. This includes modifying the engineering plans, as needed, to avoid and minimize long term impacts to surface and subsurface resources and re-establishing conditions post-construction.
- Preparing and implementing dewatering practices to prevent sedimentation into waterways.

- Avoiding the withdrawal of water from surface waters. If withdrawal cannot be avoided, all pumps and intakes should be floating to minimize sediment intake and be screened to minimize impacts to aquatic species.

- Preparing and implementing an invasive species management plan that identifies known areas of invasive species populations, addresses site restoration activities, and includes specific protocols to prevent and minimize the spread and introduction of invasive species.

- Marking TCSB's to alert navigators.

- Restoring waterway banks to pre-existing conditions (if pre-existing conditions are not stable, then restoring waterway banks to stable conditions).

- Isolating all soil piles from waterways with perimeter erosion control BMP's.

- Checking equipment for fluid leaks before crossing TCSB's.

- Anchoring TCSB's to prevent them from washing away during high flow conditions.

- Monitoring TCSB's daily for debris and removing debris, as necessary.

- Completing open-cut trenching in the waterway using a work zone isolation system or bypass.

- Gradually removing flow bypass structures in a manner that minimizes downstream sediment impacts.

- Scheduling construction to avoid disrupting sensitive species.

- Limiting the amount of time necessary to complete construction.

The applicant stated the following measures would be implemented as part of waterway impact minimization:

- Providing site-specific waterway crossing plans for DNR review and approval.

- Accessing the project corridor on either side of a waterway, by using existing public crossings, culverted drives, or existing ford crossings to the extent practical.

- Working with private landowners to identify alternate access routes to further reduce the use of stream crossings, if possible.

- Limiting the area of open-cut trenching to the minimum area required for the pipeline installation.

- Completely open-cut trenching and restoration within 24-48 hours to reduce the

timeframe of impacts.

- Placing flow bypass structures/work isolation structures (e.g. dam and pump and dam and flume methods) immediately before trenching within that waterway and removing immediately after construction activities and after the proper restoration protocols have been completed.

- Installing and maintaining appropriate erosion control measures where soil disturbance occurs near waterways and at temporary waterway crossings until construction disturbances are restored and conditions are permanently stabilized.

- Conducting additional investigations/evaluations prior to construction to better plan for boring installation across/adjacent to high-quality waterways.

- Avoiding grading along the banks of waterways that are proposed to be bored.

- Extending boring paths to cross multiple resources whenever possible.

- Restoring the bed and banks to preconstruction conditions as quickly as practicable.

- Avoiding mechanical clearing of trees and shrubs for waterways that are proposed to be bored within 25-feet of a regulated waterway and hand-trimming an approximate three-foot path to the waterway.

- Leaving herbaceous vegetation largely intact outside of the trench width for waterways proposed to be open-cut.

- Avoiding impacts to root structures within waterway buffers.

- Limiting trench widths to 8-feet wide

- Operating equipment used for open-cut trenching installation from the waterway banks and utilizing TCSBs to access the banks.

- Utilizing energy dissipation devices and screened and floating pumps during open-cut trenching

- Gradually removing flow bypass structures

5.5.4 Waterway Restoration

The applicant states detailed surveys to identify the specific characteristics of each waterway have been completed or would be completed. These would include topographic surveys, water depth and flow measurements, substrate makeup, and water flow features. This information would then be used to recreate the bed and bank conditions to the greatest extent practicable during restoration efforts.

The applicant verified that unless the existing substrate is silt, the originally removed material from the waterway would be used as backfill material for the dredged trench if the disturbed site is immediately stabilized. If the existing substrate is silt, the applicant would backfill with clean, washed gravel, crushed stone, or clean river stone to replace the excavated material.

The applicant states waterway bed and banks would be restored to pre-construction conditions as quickly as possible and once the project is complete and banks restored, vegetation would be allowed to regrow within these areas.

5.5.5 Wetland and Waterway Permitting

DNR participates in the joint review process with the Commission, as detailed in Wis. Stat. § 30.025, with respect to wetlands, navigable waterways, and stormwater management. Wisconsin Stat. § 30.025 describes DNR process for reviewing and permitting utility projects that require authorization from the Commission and DNR.

DNR is responsible for regulating the discharge of dredge and fill material into wetlands under Chapter 281.36, Wisconsin Statutes, and Wisconsin Administrative Code. As currently proposed, the project would require an individual wetland permit for the temporary wetland fill and the forested wetland conversion. Compensatory wetland mitigation would not be required for this project, per Wis. Stat. § 281.36 (3n)(d)2.

DNR is responsible for regulating impacts to navigable waterways and waterbodies under Chapter 30, Wisconsin Statutes, and Wisconsin Administrative Code. As currently proposed, the project would require waterway permit coverage from the DNR for the installation of TCSB's across waterways and dredging.

The USACE and/or U.S. Fish and Wildlife Service (USFWS) might also require additional permits and approvals.

CA's granted by the Commission are often contingent upon an applicant's ability to secure all necessary permits from state and federal agencies. Likewise, any permit granted by DNR or USACE would be contingent on the implementation of all mitigation procedures ordered by the Commission in its CA authorization.

5.6 DNR Properties

As currently proposed, the project would cross DNR-owned and managed properties, including trails. Project components crossing and/or impact DNR-owned and managed properties would require authorizations, permissions, and/or easements from the DNR.

Prior to beginning construction, the applicant would be required to coordinate and receive necessary easements/authorizations from DNR Real Estate for any DNR-owned or managed properties crossed and/or impacted by the project. Depending on the project, DNR property, and construction activities, the applicant may need to coordinate with DNR property managers and DNR Real Estate to implement mitigation measures to avoid or minimize impacts to DNR property and its resources, such as reseeding affected areas, creating/enhancing pollinator habitat post-

construction, and minimizing impacts to recreational users of the park. The applicant may also need to coordinate with DNR Real Estate and DNR grant/funding staff if the properties are encumbered by federal, state, local funds.

5.6.1 White River State Trail

The White River State Trail is a DNR-owned,19-mile trail that follows a former rail corridor between Elkhorn in Walworth County and Sturtevant in Racine County, with a short gap in Burlington. The trail passes by numerous bridges, scenic vistas, smaller townships, farmlands, and wetlands.

Route A and Route B are proposed to cross the White River State Trail in the Town of Dover. The trail is proposed to be crossed by the pipeline via boring, specifically, the jack and bore method. The applicant states installation of the pipeline would not result in impacts, land disturbance, or closures of the trail. The portions of the trail crossed by the project routes are not funded by the Land and Water Conservation Fund (LWCF) Program, a federal program that encourages the creation and interpretation of high-quality outdoor recreational opportunities.

A staging area (identified as Staging Area 7 by the applicant) located west of highway 75 (S. Beaumont Ave) would intersect a portion of the White River State Trail. This area is an existing driveway entrance that is actively used by Conserv FS, and agricultural cooperative.

5.6.2 Richard Bong State Recreation Area

The Richard Bong State Recreation Area, located in Kenosha County, was once designated to be an air defense fighter base. The air base was abandoned in 1959, just days before concrete was to be poured for a 12,500-foot runway. The DNR-owned property is open year-round and offers a variety of recreational opportunities to the public including, ATV/UTV use, biking, boating, hiking, swimming, camping, fishing, wildlife viewing, hunting, and a nature center.

Route C is proposed to cross the Richard Bong State Recreation Area. The pipeline installation within the property would result in land disturbance from open-cut trenching, bore pit excavation, and construction and equipment use. The wetland complex located on the property would be crossed via boring. Temporary and permanent structures, including removal of existing structures, are currently not proposed by the applicant. The applicant states the construction would not result in long-term impacts to the wetlands, waterways, wildlife, fisheries, vegetation, facilities, etc. from the project. According to the applicant, there would be no impacts to public use or access during construction.

According to the applicant's project maps, Route A is also proposed immediately adjacent to, but otherwise outside of, the Richard Bong State Recreation Area.

Construction on the property could impact the existing wildlife habitats within the wetlands and grasslands. The property is an Important Bird Area, especially for grassland songbirds. The property also supports a managed hunting program, including pheasant hunting in early October to the first week of January. On the east side of HWY 75 (248th Ave.), there are pockets of mature

oak trees (at least 70 – 90 years old) that would be impacted by the proposed work. The smaller pocket is across from the state property and the larger woodland area (six - seven acres) is approximately 1,500 feet northwest of state property. Oak regeneration can be difficult, and the clearing of these trees could have long-term impacts to wildlife habitat and aesthetics of the area.

DNR has requested the applicant commit to the following actions within the property:

- Restoring the grasslands back to grasslands via replanting of pre-existing (native) grassland species.

The applicant responded it would commit to restoring the grasslands back to pre-existing conditions and the applicant's restoration plan would include pre-existing grassland species. Coordination with DNR on possible alternatives would also be considered.

- Restoring the wetlands via replanting of pre-existing emergent wetland species.

The applicant responded it would commit to restoring the wetlands back to pre-existing conditions, however, because the wetland is proposed to be crossed via boring, no wetland impact is expected.

- Coordinate with the DNR property manager regarding work and restoration within the agricultural field the DNR currently leases to a local farmer.

The applicant responded it would commit to following their Agricultural Mitigation Plan for areas currently being farmed and would also be open to discussing other options with the DNR for possible restoration of these areas back to native prairie.

- Constructing outside of the nesting season: the property is an Important Bird Area with a focus on grassland songbirds.

The applicant responded it would commit to constructing within the DNR property outside of grassland songbird nesting seasons and they would work with the DNR to determine the appropriate dates. The applicant anticipates the songbird nesting season would be from mid-April until end of July.

- Completing brush clearing under frozen ground conditions to avoid rutting the wetland areas.

The applicant responded it would commit to this request.

- Restoring the upland shrubby-grassy area via replanting of pre-existing grassland species.

The applicant responded it would commit to restoring the upland shrubby-grassy area back to pre-existing conditions and the applicant's restoration plan would include pre-existing grassland species. Coordination with DNR on possible alternatives would also be considered.

- Constructing outside of pheasant hunting season (early October to first week January).

The applicant responded it would commit to coordinating with the DNR on the optimal time to complete construction, taking into account hunting season within the property.

Along HWY 75 (CTY DD) and the Richard Bong State Recreation Area, there is a ridge that is approximately 10-15 feet above the road. The applicant states the ridge would not be permanently altered upon pipeline installation, and that once the pipeline installation is complete, the ridge would be restored to its original grade and elevation. Restoration would include backfilling with the original spoil, final grading to match preexisting conditions, and applying appropriate seed and stabilization measures.

The portions of the property that would be crossed and impacted by Route C is funded by LWCF and may also require approvals from the National Park Service.

5.7 Endangered Resources

The state's Endangered Species Law, Wis. Stat. § 29.604, makes it illegal to take, transport, possess, process, or sell any wild animal that is included on the Wisconsin Endangered and Threatened Species List. In addition, it is illegal to remove, transport, carry away, cut, root up, sever, injure or destroy a wild plant on the Wisconsin Endangered and Threatened Species List on public lands. Although utility practices are exempted from the taking prohibitions of listed plant species on public lands, it may still be prudent for the applicant to actively avoid activities in certain areas that are known to host rare plants. The Federal Endangered Species Act (ESA) protects all federally listed animals from direct killing, taking, or other activities that may be detrimental to the species. Federally listed plants have similar protection, but the direct killing or taking prohibitions are limited to federal lands or when federal funds/permits are necessary. In addition, there may be other state and federal laws protecting rare species including the federal Migratory Bird Treaty Act, the federal Bald and Golden Eagle Protection Act, and the state Protected Wild Animals (Wis. Admin. Code § NR 10.02).

"Endangered resources" is a term that includes endangered, threatened, and special concern species, as well as certain natural communities and animal concentration sites. "Endangered" means a species is in danger of extinction throughout all or a significant portion of its range. "Threatened" means a species is likely to become endangered within the foreseeable future. At the state level, "Special Concern" refers to those species where some problem of abundance or distribution is suspected but not yet proved. The main purpose of this category is to focus attention on certain species before they become threatened or endangered. Reviews were completed at both the state and federal level to determine what, if any, actions may be required or recommended to avoid and/or minimize impacts to federal and state endangered resources. A summary of the endangered resources review is provided below.

5.7.1 Federally-listed Endangered Resources

Portions of this project (Routes A and B) are within the Rusty Patched Bumble Bee High Potential Zone (RPBB HPZ). Follow-up actions when within the HPZ and when suitable habitat is present may include (as applicable):

- use native trees, shrubs and flowering plants in landscaping,
- provide plants that bloom from spring through fall (Wisconsin Native Plant Species List),
- remove and control invasive plants in any habitat used for foraging, nesting, or overwintering,
- where pollinator habitat is present, complete work during the inactive season from mid-October through mid-March.

No other federally listed species came up within the state ER Review. The applicant should work with the U.S. Fish and Wildlife Service to ensure compliance with federal endangered species laws.

5.7.2 State-listed Endangered Resources

A Certified Endangered Resources (ER) Review was completed for the proposed project (ER Log #24-541) on November 5, 2024. The ER Review is based off information from the Natural Heritage Inventory (NHI) database, maintained by the DNR's Bureau of Natural Heritage Conservation. The purpose of the ER Review is to use the NHI database to identify any known endangered, threatened, or special concern species, natural communities, or animal concentration sites within and near the proposed project area. The construction area and buffers (one-mile for terrestrial and wetland species, two-miles for aquatic species) are reviewed for any endangered resources records. The applicant completed a draft ER review for the proposed project, then sent it to the DNR for verification. The ER review was checked, modified (if needed), and approved by the Endangered Resources Utility/Transportation Liaison. The NHI database contains known records for endangered resources. However, most areas of the state have not been surveyed extensively or recently, so the NHI data should not be solely relied upon, particularly in areas dominated by private lands. In areas where suitable habitat exists for protected species, but occurrences have not been recorded in the NHI database, there may be recommended activities that could mitigate or avoid potential impacts to protected species.

If approved, this project may begin construction over a year from the Certified ER Review date. DNR regularly updates the NHI database as new species records are discovered or when known populations are updated. Also, any species delisted would be removed from the database. If the project is approved, the applicants should conduct an updated review closer to the construction start date to determine if any changes to the ER Review would be needed. An ER Review should also be completed annually for ongoing maintenance and vegetation management activities.

The ER Review for the proposed project found 44 species, animal concentration sites, and natural communities within the search buffers of all work areas. As Route C is mixed in with

Routes A and B, it should be noted that all the species below also pertain to the various Route C options. Based on this information as well as information available from the DNR, the project area, and planned activities as described in the application, this project would have impacts on endangered or threatened species. Of those endangered resources where suitable habitat is present, those impacts can be minimized or even avoided if the following actions are implemented:

- One state threatened bird species may be present along Route A and B and may be impacted by project activities. There are also three special concern bird species, two of which may be present along Route B and one which may be present along Route A. The applicant has stated[11] that they would be completing presence/absence surveys in 2025 for the state threatened species. If absent, then no further actions are required. If found to be present or if assuming presence, then they would avoid all impacts to suitable habitat during the species' nesting season. These same measures would be recommended for the special concern bird species.

- Two upland and three wetland natural communities are known to be present within the project area or immediately adjacent to the project area along Route B. Natural communities may contain rare or declining species, and their protection should be incorporated into the project design as much as possible. Therefore, it is recommended to minimize impacts which may be implemented by following invasive species BMPs and/or conducting work under frozen ground conditions when working within or adjacent to this natural community.

- A special concern mammal may be present along Routes A and B. Impacts should be minimized to the extent practicable when working within suitable habitat, especially during the summer and winter months when the species is most vulnerable.

- A total of six state special concern, three threatened, and four endangered plant species may have suitable habitat that would be impacted along project area. Specifically, 11 of these species may be found along Route A and eight of these species may be found along Route B. Avoidance and minimization efforts may include 1) site surveys to confirm presence/absence of species and avoid areas of occupied habitat, or 2) conducting work above ground (avoiding impacts to the root system) during winter or under frozen ground conditions.

- A special concern and federally listed bumble bee is likely present and the applicant should follow measures set forth by the USFWS.

- A special concern herptile may be present along Routes A and B and recommended actions for this species include timing restrictions when working within suitable habitat and/or installing herp fencing.

[11] Response to DNR-MH-3.15

In addition to what was listed in the ER Review, the following are additional recommendations the applicant could consider to further ensure rare and non-game species impacts are minimized.

- The applicant should request a renewal of their ER Review closer to the construction start date (no more than one year out) to determine if any changes need to be made to avoid impacts to protected species.

- Tree clearing should be avoided during the general bird nesting and bat maternity roost season from April 1 – August 15th. The applicant has agreed to not clear trees from April 15 – July 1 for oak wilt but extending that timeframe would benefit birds and bats as well.

Based on the information available from the DNR, the proposed project routes, and planned activities as described in the application, this project is not expected to have a significant impact on state-listed endangered or threatened species and both routes appear to have similar impacts. However, with the significant number of plants that may be present along both Routes, those actions to minimize or avoid impacts are highly recommended.

5.8 Agriculture

Most of the agricultural land within the proposed routes appears to be under row-crop production comprised of corn and soybeans, with hay fields located along all three routes (A, B, and C) to a lesser extent. Excluding staging areas, Route A crosses 239.1 acres of agricultural land, Route B crosses approximately 246.4 acres, including 0.3 acres of specialty crop land, and Route C crosses approximately 69.1 acres. The staging areas contain 30.3 acres of cropland. Route A would cross about 236 acres of prime farmland, Route B would cross about 245 acres, and Route C would cross about 69 acres. Staging areas include about 30 acres of prime farmland. These numbers include farmland of statewide importance, prime farmland if drained, and prime farmland if drained and protected from flooding.[12] Additionally, Route A would contain 113.2 acres of permanent impacts, Route B would have 113.3 acres, and Route C would contain 32.6 acres. There would be no permanent impacts in specialty crop land.

The Wisconsin Farmland Preservation Program helps farmers and local governments preserve farmland, protect soil and water, and minimize land use conflicts.[13] This is achieved through developing farmland preservation plans and preservation zoning districts, and meeting soil and water conservation standards. Twelve parcels enrolled in the Farmland Preservation Program are intersected by the proposed project. Two of these parcels are intersected by multiple segments. These twelve parcels total 612.8 acres, of which 41.2 acres intersect the project area.

Drainage districts are local governmental entities formed for the purpose of draining lands for agriculture, which is done so by managing drains.[14] Constructed drains are necessary for farms to remove excess water from their land in order to grow crops. Numerous drainage districts would be intersected by the proposed project, as shown in Appendix F Attachment 21 of the

[12] PSC REF#: 537754 Response-Data Request-PSC-Guagliardo-EAT-5.01 Attach 01
[13] Department of Agricultural, Trade, and Consumer Protection. Farmland Preservation Program.
[14] Department of Agricultural, Trade, and Consumer Protection. Drainage Districts.

application.[15] The applicant stated that the County Drainage Board would be notified before undertaking any action, including any change in land use that would alter the flow of water into or from a district drain, increase the amount of soil erosion or the movement of sediment solids to a district drain, affect the operation of the drainage district, or the costs incurred by the drainage district.

The Wisconsin Department of Agriculture, Trade, and Consumer Protection (DATCP) has produced an Agricultural Impact Statement (AIS) to investigate impacts to agricultural landowners.[16] This information is useful for landowners in the easement negotiation process and would be provided to landowners as part of that process. The AIS recommended to the PSC that WE-GO hire an Independent Environmental Monitor (IEM) or Independent Agricultural Monitor (IAM) for the duration of project construction and that all reports generated by the IEM/IAM should be shared with the PSC, DATCP, and DNR. The AIS further recommended that the IEM/IAM should be required to complete DATCP's standard Agricultural Monitoring Form for Pipeline Projects, as seen in Appendix H of the AIS and provide monitoring forms. A full list of recommendations from DATCP to the PSC and WE-GO is provided in the AIS.

After construction is completed, the project may continue to affect agricultural productivity for years afterwards. Yield reductions can be caused by inadequate protection of topsoil, changes to surface and subsurface drainage, construction debris left in fields, and opportunistic weed growth. Where construction equipment crosses cropland, the soils are likely to be compacted. Utilities can choose to decompact the soils themselves or pay the farmer to either restore the soils themselves or hire a contractor to do the work for them. Proper restoration of the compacted soil is necessary for crop yields to return to pre-construction levels. Even so, sometimes it may take several years for the soil health to return. These and other problems can increase costs for the farm operators. WE-GO indicated that landowners could contact the company to investigate any potential mitigations for crop yield loss identified beyond five years.[17]

Research shows that pipeline installations have long-term negative effects on soils and plants. A literature review containing 34 studies from eight countries that reported pipeline impacts on agricultural and natural ecosystems concluded that soil degradation occurred after pipeline installation due to increased compaction, increased soil mixing, decreased aggregate stability, decreased water infiltration, and decreased soil carbon. This degradation led to the decline of plant productivity, with 15 out of 25 studies recording a decline in crop yields.[18] The researchers concluded that pipeline installations typically result in degraded soils and crop yield losses, which can persist for decades following installation. Another study found the impact of pipeline installation on soils and field crops persisted for five years following pipeline installation and that current best management practices were insufficient to combat widespread soil degradation and crop yield loss.[19]

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[15] PSC REF#: 519126 Appendix F Attachment 21 Drainage Districts Table and Map
[16] PSC REF#: 538519 Revised Agricultural Impact Statement #4622
[17] PSC REF#: 535460 Response-Data Request-PSC-Thompson-EAT-3.05
[18] Brehm and Culman. 2022. Pipeline installation effects on soils and plants: A review and quantitative synthesis. Agrosystems, Geosciences & Environment.
[19] Brehm and Culman. 2022. Soil degradation and crop yield declines persist five years after pipeline installations. Soil Science Society of American Journal.

WE-GO developed an Agricultural Mitigation Plan (AMP)[20] and included it in the project application. The purpose of the AMP is to describe agricultural construction mitigation and restoration methods to be used on the project, establish personalized communication with agricultural landowners, provide agricultural landowners and tenants with a hotline to contact the company Representative, and describe the duties of the WE-GO Agricultural Inspector. The Agricultural Inspector would have familiarity with agricultural operations, general construction, and knowledge of agronomy and soil conservation.[21]

Trench depth in agricultural lands would be sufficiently deep enough to allow a minimum of four feet of cover over the top of the pipeline to avoid possible interference with farming equipment. Material excavated during trenching in agricultural lands would have topsoil and subsoil separated, if applicable, as to not impede future growing seasons and promote healthy soil after restoration.

The applicant could minimize construction impacts on agricultural soils by using one or more of the following techniques:

- Completing construction during dry or frozen conditions;
- Distributing axle loads over a larger surface area to reduce the bearing pressure on agricultural soils;
- Identifying site specific soil characteristics and concerns from the landowner and farm operator before construction begins;
- Avoiding areas where impacts might occur by altering access routes to the construction sites;
- Using existing roads or lanes utilized by the landowner;
- Using construction mats, ice roads, or low ground pressure or tracked equipment to minimize compaction, soil mixing, rutting, or damage to drainage systems;
- Segregating topsoil or soil horizons during excavation and construction to minimize soil mixing;
- De-compacting soils following construction with appropriate equipment and when moisture levels allow for successful restoration efforts until the degree of soil compaction levels in the ROW is similar to soils off the ROW;
- Avoiding construction and maintenance activities during times when soils are saturated; and/or

[20] PSC REF#: 519132 Appendix A Attachment 20 Agricultural Mitigation Plan.
[21] PSC REF#: 535459 Response-Data Request-PSC-Thompson-EAT-3.04.

- Avoiding the removal of critical windbreaks and replanting windbreaks with lower growing woody species to minimize soil erosion due to wind.

Subsoils are less productive than topsoil and mixing the soil types should be avoided as much as practicable. This includes avoiding creating large ruts, which can lead to soil mixing. The AMP includes similar best management practices including, but not limited to, segregating topsoil, using existing farm roads for access where possible, implementing erosion control measures, working with landowners to identify irrigation systems and drain tiles to avoid damage, and restoring soil.

5.9 Archaeological impacts

In February and September 2024, Cultural Resource Management at the University of Wisconsin-Milwaukee conducted cultural resource investigations for the Rochester Lateral Project on behalf of We Energies to address responsibilities for identifying, recording, and managing cultural resources as stipulated under Wis. Stat. §§ 44.40 and 157.70.

Compliance with Wis. Stat. § 44.40

The Commission is required to request the Wisconsin State Historic Preservation Office (SHPO) to review and comment if a Commission action would affect a historic property, as stated in Wis. Stat. § 44.40 and the PSC-SHPO Interagency Programmatic Agreement (PSC-SHPO Agreement or PA). According to the Agreement, Appendix of PSC Authorization Actions Subject to Wis. Stat. § 44.40, this project is listed as Type II(a) Natural gas pipeline siting and construction. Therefore, Commission authorization of the project must comply with Wis. Stat. § 44.40.

Commission Review of WHPD Properties

The Commission's Historic Preservation Officer (HPO) reviewed and evaluated the project application materials related to historic properties, which were provided in accordance with the PSC Application Filing Requirements. The Commission HPO also reviewed and evaluated potential historic property records using the Wisconsin Historic Preservation Database (WHPD or database) online portal and its associated GIS data. As stated in the PSC-SHPO Agreement (3), the WHPD contains all listed property, the Wisconsin inventory of historic places, and the list of locally designated historic places. These recorded properties comprise all relevant "historic properties" for the purpose of this review.

Area of Potential Effect (APE)
The APE is the area where WHPD properties may be affected by the proposed activity. The PSC-SHPO Agreement (7g), requires the Commission HPO to determine the APE. The APE is the geographic area or areas within which an undertaking may directly or indirectly cause changes in the character or use of historic properties, if any such properties exist. The scale and nature of the undertaking influence the shape and extent of the APE, resulting in delineated areas of effects that may be different depending on the kinds of effects caused by the undertaking. The direct APE is typically defined as the area where physical disturbance occurs (e.g., excavation, soil grading, compression of soils through heavy machinery movement and material staging). The indirect APE is usually determined as the distance from the project where visual disturbance reasonably occurs (e.g. line of sight).

WHPD Archaeological Site Inventory (ASI) Properties in the APE
The WHPD records indicate 11 ASI properties in the APE. Archival and literature research completed by the consultant determined that seven of these WHPD ASI properties lack the materials and integrity to qualify as historic properties under Wis. Stat. § 44.36(2): KN-0241, MI-0366, MI-0513, MI-0515, MI-0542, RA-0154, and RA-0255. The consultant completed Phase I archaeological survey for RA-0145 and RA-0181, which resulted in no cultural materials or evidence of the sites in the APE.

No field survey has been completed for MI-0253 due to lack of permission from the landowner. As a result, this WHPD ASI property has not been evaluated to determine whether it meets the Wis. Stat. § 44.36(2) historic property criteria. The applicant stated that if the project is authorized, it would perform field surveys to determine whether MI-0253 would meet the historic property criteria.

WHPD ASI property BRA-0025 is a marked cemetery and permission to disturb (PTD) a human burial site is required under Wis. Stat. § 157.70 if the project would affect the cemetery. The consultant's report states that unmarked burials associated with a cholera epidemic are reported in the southern part of the cemetery and may extend further, with previous recommendations to monitor the area during construction, specifically north of Olson Road where the cholera burials are reportedly located. The consultant performed visual investigations at BRA-0025, which concluded that marked burials lie within the right of way. The consultant photo documented the distance between the nearest burial markers and the road edge and confirmed that graves extend into the ROW. They noted that a ROW marker in the southeast part of the cemetery measures 5.1m from the road edge, two headstones are 4.09m from the road edge of Highway 45, and six other headstones are between 4.7m and 5.15m from the road edge of Highway 45. Should the project intersect the WHPS ASI property boundary for BRA-0025, then the applicant would require PTD to comply with Wis. Stat. § 157.70.

Cemeteries, graves, and religious properties are not typically able to qualify as historic properties, unless they are an integral part of a district that meets the criteria of significance under Wis. Stat. § 36(2)(a) or unless they are a religious property deriving primary significance from architectural or artistic distinction or historical importance. None of these criteria have been met, therefore, the site would not be considered a historic property for the purposes of Wis. Stat. § 44.40.

WHPD Architecture History Inventory (AHI) Properties in the APE

The proposed pipeline would be constructed underground, as a result the APE for above-ground historic properties would be limited to any ground disturbance or new above-ground facilities within visual range. The WHPD records identified 21 AHI properties in the APE. Of these, the George Hardie Farmstead (12334, 63585, 63586, 63587) is the only property that was formally determined eligible for the National Register of Historic Places (NRHP) and therefore meets the criteria to be considered a historic property under Wis. Stat. § 44.36(2). The applicant stated that there would be no above-ground facilities that would have a visual effect on the George Hardie Farmstead. One property, Mahr's Hillside Farm and Orchard (147128, 147131, 147132, 147133), was formally determined as not eligible as a historic property. WHPD records for the remaining 19 resources indicated most had been previously surveyed and the consultant believes they would not meet historic property criteria and would be unlikely to be affected by the underground project facilities.

Conclusion and HPO Determination

The Commission HPO reviewed the project for affects to historic properties in accordance with the PSC-SHPO Agreement. There are no known historic properties in the APE that would be affected by the project and the applicant states that they would do additional survey for MI-0253 if the project were approved and a route selected in that area. Therefore, the Commission is not currently required to request SHPO review and comment on this undertaking as required by Wis. Stat. § 44.40.

5.10 Contaminated Sites

A review of the Wisconsin Remediation and Redevelopment Database was completed for an area within two miles of the project. Four open sites are associated with Route A, seven are associated with Route B and eight are associated with both Routes A and B. Twenty-eight closed sites are associated with Route A, 48 are associated with Route B and 83 are associated with both Routes A and B. Collectively within two miles the of the proposed project corridors, there are 19 open and 159 closed contaminated sites within a two-mile radius of the project area. Based on the nature and locations of the open and closed contaminated sites, further evaluation is necessary to determine if contamination may be encountered during construction.

A review of the Wisconsin Historic Registry of Waste Disposal Sites was completed for an area within two miles of the project. There are 27 sites located within a two-mile radius of the project area (seven along Route A; seven along Route B, and 13 associated with both Routes A and B). Based on the distance of these sites to the project area, further evaluation is necessary to determine if contamination may be encountered during construction.

If the project is approved, the applicant would evaluate the contaminated sites located within the project ROW and prepare a plan to address potential soil or groundwater contamination during construction.

5.11 Parks and Recreational Areas

The proposed project intersects parks and recreational trails at various locations, as shown in the table below. Route A intersects two parks and two trails, including Gorney Park and White River State Trail; Segment AB (associated with both Routes A and B) intersects two sections of the Route of the Badger trail; Route B intersects two parks and the White River State Trail; and Route C intersects Beaumont Park, Bong State Recreation Area, and Brighton Dale Park.

Table 10: Parks and Recreational Areas Trails within the Project Area

Segment	Park/Trail Name	Owner
A-6	Gorney Park	Village of Caledonia
A-7	Unnamed park including the Root River	Racine County
A-7	Unnamed trail	Caledonia Conservancy Ltd.
A-10	White River State Trail	DNR
AB-3	Route of the Badger	WI Electric Power Co.
AB-3	Route of the Badger	WI Electric Power Co.
B-7	Unnamed park including the Root River	Racine County
B-7	Unnamed park including the Root River	Racine County
B-11	White River State Trail	DNR
C-4	Beaumont Park	Racine County
C-11	Bong State Recreation Area	DNR
C-11	Brighton Dale Park	Kenosha County

WE-GO has met with the DNR and Kenosha County to discuss temporary and/or permanent easements within recreational areas. In a data request response,[22] the applicant stated that the White River State Trail crossing would be completed using the jack and bored method, resulting in no impacts, land disturbances, or closures of the trail. The applicant also stated that land disturbing activities in Bong State Recreation Area would include trenching and HDD within the easements.

During construction, timber matting and other BMPs would be used to minimize impacts to resources near or within any park or recreational area. Existing ROW easements would also be used where possible to avoid any unnecessary clearing. Once construction activities are complete, the matting would be removed, and the ground surface restored to pre-existing conditions to the extent practicable.

To ensure public safety, portions of these areas may need to be closed temporarily during construction. The company would minimize the area and duration of closures as much as possible and work with the land managers to communicate any necessary closures to park and recreational area users.

[22] PSC REF#: 524790 Response-Data Request-PSC-Thompson-WDNR-MH-1.4

5.12 Conservation Easements

WE-GO identified one conservation easement that intersects the proposed pipeline along Segment B-6. This easement is currently enrolled in the wetlands reserve program. The company stated that implementation of HDD construction methods would be used wherever possible to avoid impacts to this and other related properties.

5.13 Air Emissions and Climate Change

This gas pipeline project would contribute to greenhouse gas emissions and climate change some extent through natural gas extraction, distribution, and combustion processes.

While natural gas may burn cleaner than alternative fuels including coal and petroleum, the extraction and delivery of natural gas and the operation of natural gas power plants has meaningful contributions to US greenhouse gas (GHG) emissions, particularly because of methane. Methane (CH_4) is the main component of natural gas and has 28 times the potency of carbon dioxide (CO_2) as a GHG, meaning it is much more efficient at trapping heat.[23] Other components of natural gas include natural gas liquids and nonhydrocarbon gases, such as carbon dioxide and water vapor. Natural gas liquids are a group of hydrocarbons including ethane, propane, normal butane, isobutane, and natural gasoline.[24]

In 2022, methane constituted 11.1 percent of the reported US GHG emissions, and natural gas systems were the second largest source of methane emissions following agricultural sources[25]. For every 1,000 kg (2,204 lbs) of natural gas delivered to a power plant, 12.5 kg (27.5 lbs) of methane is released to the atmosphere, which is equivalent to a 1.1 percent loss of natural gas extracted.[26] Within the oil and natural gas industry, production of gas and oil accounts for 60 percent of total methane emissions and transmission and storage account for 19 percent of methane emissions. Within the transmission segment, pipeline venting accounts for nine percent of methane emissions.[27]

While reported natural gas emissions have reduced dramatically from the 1990's, methane emissions have come under especially high scrutiny due to their large global warming potential. Some of the most recent life cycle research stemming from a collaboration at Brown University, Rocky Mountain Institute, Harvard University, NASA, and Duke University concluded that global gas systems that leak over 4.7 percent of their methane (20-year timeframe) or 7.6 percent (100-year timeframe) have life cycle emission intensities that are equivalent to coal emissions[28]. Another study found that methane losses in gas systems must be kept below 3.2 percent for natural gas power plants to have lower life cycle emissions compared to new coal plants over

[23] US Environmental Protection Agency. Last updated January 7, 2025. Overview of Greenhouse Gases.
[24] US Energy Information Administration. Last updated October 10, 2024. Natural gas explained.
[25] US Environmental Protection Agency. Last updated November 22, 2024. Inventory of U.S. Greenhouse Gas Emissions and Sinks.
[26] Skone, et. al. 2014. Life Cycle Analysis of Natural Gas Extraction and Power Generation. U.S. Department of Energy National Energy Technology Laboratory.
[27] US Environmental Protection Agency. Last updated on June 18, 2024. Estimates of Methane Emissions by Segment in the United States.
[28] Gordon, et. al. 2023. Evaluating net life-cycle greenhouse gas emissions intensities from gas and coal at varying methane leakage rates. Environmental Research Letters, 18(8), 084008.

20 years or fewer.[29] According to an analysis by the Environmental Defense Fund (EDF), natural gas pipelines nationwide leak as much as 2.6 million tons of methane each year, which has the same climate impact as nearly 50 million passenger cars for a year on near-term warming scale. EDF also estimates that US onshore gas pipeline leaks are 3.75 to eight times greater than what is estimated by the EPA.[30] An analysis of gas leak incidents based on US Department of Transportation, Pipeline, and Hazardous Materials Safety Administration data found that a major gas leak incident occurs once every 40 hours in the US[31].

Extraction of natural gas may also include fracking, which is a process that contributes to various other environmental issues such as contamination of air and water, disruption to land and local ecosystems, and consumption of large volumes of water.[32,33,34] A complete assessment of all associated GHG emissions and environmental impacts from all life cycles of natural gas is beyond the scope of this analysis. The overall natural gas and fuel production and transportation system is very complex and is impacted by many more factors than this proposed project alone.

Potential impacts of GHG emissions on global climate change and the potential effects of climate change are described in reports of the Intergovernmental Panel on Climate Change, the scientific body set up by the World Meteorological Organization and the United Nations Environment Programme to provide information about global climate change.[35] Governor Evers' Executive Order 52[36] directed the Wisconsin Initiative on Climate Change Impacts (WICCI) to collect and update scientific data on the rate of climate change in Wisconsin and its impact on Wisconsin's natural environment. Executive Order 38[37] also contained a goal for all electricity consumed within the state of Wisconsin to be carbon-free by 2050. The electric grid and system are highly complex, but if the projected life of this proposed pipeline would extend past 2050, then this project could conflict with this goal to some extent. However, this particular project is a small part of the overall state goal, and it is unknown what the exact electric generating mix will be in 2050.

[29] Alvarez, et. al. 2012. Greater focus needed on methane leakage from natural gas infrastructure. Environmental Sciences.
[30] McVay. 2023. Methane Emissions from U.S. Gas Pipeline Leaks. Environmental Defense Fund.
[31] Dutzik et al. 2022. Methane Gas Leaks. U.S. Public Interest Research Groups Educational Fund.
[32] Meng. 2017. The impacts of fracking on the environment: A total environmental study paradigm. Science of The Total Environment.
[33] Mehany and Guggemos. 2015. A literature survey on the fracking economic and environmental implications in the United States. Procedia Engineering.
[34] Environmental Protection Agency. Last updated on December 30, 2016. Natural gas extraction – hydraulic fracturing.
[35] For example, the website of the Intergovernmental Panel on Climate Change provides a source of information and reports about global climate change.
[36] Wis. Governor's Office, Executive Order #52: Relating to the Creation of the Governor's Task Force on Climate Change. 2019.
[37] Wis. Governor's Office, Executive Order #38: Relating to Clean Energy in Wisconsin. 2019.

Direct, indirect, and cumulative effects of fossil fuel combustion have been shown to impact human health and mortality worldwide.[38],[39],[40],[41] Research shows that air pollution from fossil fuel combustion is the leading environmental threat to children's health and equity.[42] Similarly, the World Health Organization (WHO) states that air pollution is one of the greatest environmental risks to child health and is estimated to have caused 4.2 million premature deaths worldwide in 2019.[43] The December 2020 report produced by Governor Evers's Task Force on Climate Change summarizes the human and economic costs of climate change on Wisconsin communities and emphasizes that the potential climate change impacts in Wisconsin would disproportionately impact Black, Indigenous, and other communities of color and low-income communities.[44] The WICCI also reported on how Wisconsin's climate continues to change and the impacts of our warming climate on Wisconsin residents.[45]

5.14 Invasive Species and Disease Organisms

During a 2024 field visit along the project routes, 14 invasive plant species were identified. The project also occurs within the known range of the emerald ash borer and the spongy moth and is within counties where oak wilt is known to occur.

Construction activities may allow for or cause the spread and establishment of invasive species. Construction equipment traveling from infested to non-infested areas could spread noxious or invasive weed seeds and propagules. The removal of existing vegetation and disturbance of soils during construction could create conditions that allow the spread and establishment of noxious and invasive weeds. Generally, areas with established populations of invasive plants can include field edges, road ROWs, wetlands and waterways, and potentially fallow fields or areas not in active management.

There are a range of actions that can be taken before, during, and after construction to prevent the introduction and spread of invasive species. The Wisconsin Council on Forestry has BMPs[46] on controlling invasive species for a range of industries and activities, including for transportation and utility rights-of-way. The use of these BMPs can assist in complying with the "reasonable precaution" requirements under NR 40. The applicant states that during construction of the project, the potential to introduce or spread invasive species would be mitigated in compliance with requirements of NR 40. The DNR Harvesting Guidelines to Reduce the Risk of Introduction and Spread of Oak Wilt would be followed to minimize the

[38] Pozzer, et. al. 2022. Mortality Attributable to Ambient Air Pollution: A Review of Global Estimates. GeoHealth.
[39] Murray, et. al. 2020. Global burden of 87 risk factors in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. Lancet.
[40] Schraufnagel, et. al. 2019. Air Pollution and Noncommunicable Diseases: A Review by the Forum of International Respiratory Societies' Environmental Committee, Part 2: Air Pollution and Organ Systems. Chest.
[41] National Institute of Environmental Health Sciences. August 2020. Air Pollution and Your Health.
[42] Perera. 2017. Pollution from Fossil-Fuel Combustion is the Leading Environmental Threat to Global Pediatric Health and Equity: Solutions Exist. International Journal of Environmental Research and Public Health.
[43] World Health Organization. 2024. Ambient (outdoor) air pollution.
[44] Governor's Task Force on Climate Change Report. December 2020, pp. 14-17.
[45] University of Wisconsin-Madison. Wisconsin Initiative on Climate Change Impacts.
[46] Wisconsin Council on Forestry. Invasive Species Best Management Practices for Transportation and Utility Rights-of-Way. Version: January 6, 2010.

potential to spread oak wilt, and the applicant would also follow the state's (DATCP and DNR) regulations and requirements to reduce the potential to spread emerald ash borer. The applicant states that some of the planned actions to control the introduction and spread of invasive species during construction of this project include:

- Avoidance through construction timing and alternate access
- Proper management of construction vehicles and materials, including timber matting (i.e., storage, cleaning)
- Minimizing ground disturbance
- Placing a barrier between construction vehicles and plants (i.e., timber matting)
- Proper storage and disposal of plant material
- Promoting native regeneration
- Leaving cut vegetation onsite where it is cut (i.e., mowing shrubs)

These actions are consistent with some of the recommended BMPs discussed in the WI Council on Forestry document. If these and other BMPs are followed during construction activities, there is not likely to be a significant increase in invasive species populations in the area.

6. Community Impacts

WE-GO has sent public outreach mailings and handouts that describe the project and has had ongoing discussions through individual meetings/phone calls with local officials and impacted landowners. The company continues discussions regarding the scope of the project with landowners and the following:

- Village of Caledonia
- Town of Raymond
- Town of Norway
- Town of Dover
- Town of Brighton
- Village of Rochester
- Racine County
- Kenosha County
- City of Oak Creek

Invitations to public information meetings were sent to all landowners within the environmental survey area, which is typically 250 to 300 feet on each side of the proposed routes. Meetings were held on March 4, 2024, in Raymond, WI; March 7, 2024, in Kansasville, WI; and March 13, 2024, in Caledonia, WI. Approximately 100 people attended the three meetings and individuals reviewed proposed route maps, asked questions of company representatives and received information about the proposed project.

Property owners would be notified of the upcoming work a few weeks prior to start of construction through in-person contact, phone call and/or a letter process. They would be provided a contact to call if they have any concerns regarding the project. If access to a property is required, individual property owners would be contacted by a We Energies representative in advance of accessing the property or of any work being performed. Property owners would be provided maximum allowable driveway access during construction. If a customer's driveway or access needs to be temporarily disturbed, they would be notified in advance. Temporary driveway access would be provided with road plates following excavation and pipe installation until full restoration is complete. Any customer landscaping disturbed would be restored or compensated accordingly as allowed to retain access for future maintenance to the pipeline. The company indicated that trees on private property within the easements would need to be cleared and any fencing within easements would need to be temporarily taken down. Trees would not be replanted, but as part of negotiations, landowners may be reimbursed for cleared trees.[47]

[47] PSC REF#: 535456 Response-Data Request-PSC-Thompson-EAT-3.01

Significant trees may be avoided in some circumstances, which is determined on a case-by-case basis. Through negotiations with the landowner, the company would either re-establish private fences after construction or compensate the landowner to have it re-established.[48] Any impact to sidewalks or curbs would be restored to guidelines maintained by the permitting authority. Excavations in these areas would be protected with plywood, plating or snow fencing to maintain pedestrian access where permissible. Where not permissible, an alternate route would be assigned.

The project would create increased dust in the areas of construction. Dust may be generated from excavation work, exposed soils, or materials transport, and could create a nuisance for local homeowners or drivers. The extent to which dust would be generated during construction would depend on the level of construction activity, weather conditions such as high winds, as well as the moisture content and texture of soils being disturbed. Efforts to minimize dust, such as wet brooming and watering, would be implemented when necessary. Air quality impacts during construction of the pipeline are expected to be localized and limited in duration to the use of machinery or increased heavy traffic. Diesel exhaust from construction machinery or trucks may occur in and around the work areas. These impacts are expected to be short term and would end when local construction activities are complete.

To minimize erosion and prevent sediment runoff from entering nearby residential or municipal water systems, erosion control measures would need to be implemented that comply with an approved Erosion Control and Storm Water Management Plan. Restoration would take place along the route as areas would be fertilized, seeded, and mulched with erosion mats installed to encourage the reestablishment of vegetation.

As with any construction project, the potential exists for spills of pollutants such as diesel fuel, hydraulic fluid, drilling fluids, lubricants, and solvents. Care would need to be taken when handling any potential pollutant especially during construction of the pipeline and when refueling or performing maintenance of equipment and vehicles. Spill kits and staff training in the use of these materials would decrease the risk of spills leading to site or water contamination. Solid waste would be generated during the construction of this project and would need to be removed to appropriate waste disposal or treatment facilities. Any contaminated soils found during the construction process would need to be disposed of at an approved disposal or treatment facility for those materials.

The following table shows how many times the pipeline comes within a certain distance of various building types. Segment B-6 has the highest occurrence of coming within 100 feet of a residential building, which is 11 times. Segment B-6 also has the highest occurrence of coming within 550 feet of a residential building, which is 123 times. Segment A-6 comes within 550 feet of a residential building 97 times. Segments A-6 and B-6 are also the longest segments of each route, at 15.8 miles long and 16.1 miles long, respectively.

[48] PSC REF#: 534107 Response-Data Request-PSC-Thompson-EAT-3.06

Table 11: Distances to Various Building Types

Building Type	Route A	Route B	Route C
Residential – within 100 feet	11	21	3
Residential – within 550 feet	167	188	34
Commercial – within 100 feet	0	0	0
Commercial – within 550 feet	8	10	1
Institutional – within 100 feet	0	0	0
Institutional – within 550 feet	2	0	0
Out Buildings, Shed, Garages, Barns – within 100 feet	18	23	7
Out Buildings, Sheds, Garages, Barns – within 550 feet	195	215	51

6.1 Noise

Noise is unwanted sound considered unpleasant, loud, or disruptive to hearing. Noise is measured in units of decibels (dB) on a logarithmic scale. Because the human ear is not equally sensitive to sounds throughout the range of hearing frequencies, a weighted scale is commonly used, with the A weighted scale (dBA) most often used for sound measurements affecting human hearing. Due to the logarithmic scale of sound measurements, a change of three dBA is considered barely perceptible, while a change of 10 dBA is perceived as a doubling/halving of noise. For reference, the sound level of normal breathing is about 10 dBA, normal conversation at three feet is about 60 dBA, and emergency vehicle sirens are about 115 dBA. Impacts associated with noise can be subjective and vary from person to person, based on factors such as loudness, time of day, frequency, or duration, and the amount of other background noise audible to the listener.

A range of equipment, vehicles, and machinery would create noise during construction of the project. Most of this would come from diesel engine-powered construction equipment such as dozers, graders, excavators, trenchers, dump trucks, drill rigs, and others. Employee and delivery traffic would increase in the area but is not anticipated to be a substantial source of increased noise. Construction noise impacts would vary with the time of day, stage of construction, and location of nearby receptors. The table below shows some of the typical noise levels at 50 feet for commonly used construction equipment.

Table 12: Average Maximum Sound Levels from Common Construction Equipment

Equipment	Sound level at 50 feet (dBA)
Dozer	86
Grader	79
Excavator	87
Flat Bed Truck	74
Pile Driver	105
Crane	79
Roller	82

In an effort to minimize noise impacts, work is to be completed only during permitted hours as allowed by the permitting authority, if applicable. Another way to mitigate noise impacts during construction is to ensure that diesel engine mufflers on machinery or equipment are kept in good working order.

6.2 Environmental Justice and Sensitive Receptors

Environmental justice is the fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income, with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies. Historically, communities of color and low-income communities have been disproportionally impacted by adverse human health and environmental effects associated with pollution and developments. The first step towards evaluating whether a project may have disproportionate impacts is by evaluating the population in a project area. For the RLP, local census data was reviewed to determine whether any identifiable groups of minority or low-income persons are in the project study areas.

Table 13: Population and Income (2023 data from census.gov).

Location	Village of Caledonia	Town of Norway	City of Oak Creek	Racine County	Kenosha County
Population	25,326	7,912	37,156	196,613	167,488
Median Household Income	$90,346	$104,667	$86,408	$72,658	$76,583
Poverty Rate	5.0 percent	5.5 percent	4.7 percent	9.6 percent	9.8 percent

Table 14: Estimated Racial or Ethnic Distributions (2023 data from census.gov).

Location	Village of Caledonia	Town of Norway	City of Oak Creek	Racine County	Kenosha County
White alone, not Hispanic or Latino	79.6 percent	92.5 percent	73.7 percent	69.3 percent	72.8 percent
Black alone	3.0 percent	0.1 percent	3.8 percent	11.8 percent	7.7 percent
American Indian & Alaska Native alone	0.2 percent	0.2 percent	0.1 percent	0.7 percent	0.8 percent
Asian alone	2.0 percent	0.6 percent	6.6 percent	1.5 percent	2.0 percent
Hispanic or Latino[a]	11.2 percent	5.0 percent	12.1 percent	15.4 percent	15.4 percent
Two or More Races	8.4 percent	3.4 percent	9.5 percent	3.2 percent	3.1 percent
Native Hawaiian & Other Pacific Islander alone	0.4 percent	0 percent	0 percent	0.1 percent	0.1 percent

*** Where (-) is shown, no census data on that group was provided due to low numbers*.
[a] *Hispanics may be of any race, so also are included in applicable race categories*

For the purposes of this analysis, a minority population consists of any geographic area in which representation is greater than the national average of 30 percent. The median household income[49] for the State of Wisconsin is $75,670 (Census 2023). The State of Wisconsin persons in poverty rate is 10.7 percent (Census, 2023). Through a review of the population details available, there are no disproportionately high minority populations or low-income populations identified near any of the gas pipeline facilities that would be adversely impacted by the proposed project. In addition, the EA finds there are no significant adverse impacts expected to occur to human health or communities, and therefore no disproportionate impacts to minority or low-income populations are anticipated.

Sensitive receptors are mainly those individuals that are very young, elderly, or infirm. Local day care facilities, schools, hospitals, and elderly care facilities could have a greater potential to be affected by construction impacts such as fugitive dust, increased noise, and increased traffic hazards. The applicant stated that the closest sensitive receptors were Lakeview Specialty Hospital and Rehab, located approximately 1,000 feet from Route A and 1,300 feet from Route B near STH 20 and Sharp Road.

[49] In 2023 dollars, from Census.gov. NRE.

6.3 Safety

Transporting natural gas by pipeline involves some risk to the public in the event of an accident and subsequent release of gas. The greatest hazard is a fire or explosion following a major pipeline rupture. Methane, the primary component of natural gas, is colorless, odorless, and tasteless. While not toxic, if breathed in high concentration, oxygen deficiency can result in serious injury or death.

Methane has an ignition temperature above 1,000º F and is flammable at concentrations between five and fifteen percent in air. Unconfined mixtures of methane in air are not explosive; however, a flammable concentration within an enclosed space in the presence of an ignition source can explode. It is buoyant at atmospheric temperatures, does not remain in water, and disperses rapidly in air.

Natural gas is odorless and colorless when it's shipped into Wisconsin via large, high-pressure, interstate pipelines. The local utility odorizes the gas prior to distributing the gas to its customers. Odorizing the gas helps identify leaks in both the distribution system and in the customers' buildings.

The federal government sets minimum safety standards for all pipelines; it also makes and enforces rules for interstate gas pipelines. The State of Wisconsin, specifically the PSC, makes and enforces rules for intrastate pipelines (within Wisconsin). It has adopted and enforces all federal pipeline safety standards for construction and operation of gas pipelines, except in instances where the state code contains stricter standards.

The PSC employs a team of pipeline safety engineers who monitor pipeline construction as it occurs and regularly inspect pipelines and pressure regulators to ensure that the natural gas system is operating at appropriate and safe pressures and is not subject to corrosion and undue hazards. The PSC's pipeline engineers cumulatively spend over 500 days per year in the field conducting inspections. In addition, they complete a comprehensive inspection of every gas utility at least once every three years, including the utility's maintenance records. State and federal pipeline engineers work cooperatively to achieve the goal of public safety.

Federal and Wisconsin safety regulations include, among other subjects, standards for pipeline materials selection, design, welds, construction of service regulators, corrosion control, testing, operation, and maintenance, as well as required clearances (based on population density), qualifications for pipeline personnel, and emergency response plans.

6.3.1 Pipeline Construction Safety

Inspectors from the pipeline safety program of the PSC would be present periodically during construction to ensure compliance with state and federal requirements for pipelines. Among other responsibilities, when monitoring the construction of new pipelines, the PSC's pipeline safety engineers ensure that:

- Required clearances from occupied buildings are met.
- Trench size and depth and fill materials meet required specifications.

- Pipeline components are made of standard and approved materials.
- Pipeline welders and fitters are trained and certified.
- Written procedures are followed during construction.
- Pipeline welds conform to code through visual inspection or x-ray technology.
- Pipeline coatings or wrappings to prevent corrosion are installed correctly.
- Odorants are added to the natural gas transported in a pipeline when warranted.
- Pipeline locations are marked at appropriate intervals and road crossings.
- ROW restoration is completed in a safe and timely manner.

This is a normal part of pipeline safety inspector duties, and a project of this magnitude would receive higher priority for inspections. Inspectors cite all observed violations and report to the Commission, which has the option to issue fines for violations not addressed. The purpose of this enforcement is to ensure the pipeline meets the minimum federal and state expectations for public safety and reliability. Additionally, utility-hired weld and construction inspectors are required to ensure the quality of work daily.

6.3.2 Pipeline Operation Safety

Ensuring pipeline safety for operating gas pipelines includes, among other duties, checking for leaks and corrosion, inspecting pressure gauges at compressor stations, and responding to customer complaints. PSC pipeline safety engineers review the operating and maintenance plans that each gas utility must follow and ensure that the plans are being followed.

The utility's long-term maintenance activities for the pipeline includes pressure monitoring, corrosion protection monitoring, leak surveys, patrols, and aerial surveys. An integrity management program dictates that the utility periodically conducts tests to determine the condition of transmission pipelines through pressure tests, inline (internal) inspection device runs, or direct assessment on exposed sections. All found leaks or anomalies must be prioritized and repairs are made under stringent requirements. An emergency response program ensures the utility personnel and relevant responders and officials are prepared for an unanticipated release of gas. At the request of first responders or affected municipalities, the utility can provide additional training in natural gas emergency response. The utility already provides similar education as part of its public awareness program and is required to maintain liaison with fire, police, and public officials as part of its emergency plan.

The public must follow rules relating to contacting Diggers Hotline prior to excavation activities. Diggers Hotline is Wisconsin's statewide one-call notification system that was developed in order to provide excavators and the general public the ability to inform multiple owners of underground facilities of excavation with a single call. Diggers Hotline is a not-for-profit independent company

that processes information from excavators and the general public and passes that information on to affected utilities. Diggers Hotline does not locate buried facilities in the field.

Wisconsin Stat. § 182.0175 requires excavators to notify Diggers Hotline before starting work to excavate, grade, trench, dig, drill, augur, tunnel, scrape, plow cable or pipe. Excavation, according to the law, means anything that moves, removes or displaces earth rock or other material in or on the ground. State law also mandates an 18-inch buffer zone on each side of a marked facility. No mechanized equipment can be used within this buffer zone; only hand digging is allowed.

6.3.3 Developed land

Route A may affect developed land, including several types of buildings and land uses. Possible impacts could include aesthetics, accessibility, future development, property values, safety, and recreational use. WE-GO provided a summary of the types of properties, identified through aerial imagery, which would be located within certain distances of the proposed Route A ROW. These include residential buildings, commercial buildings, institutional buildings, and outbuildings.

The company's preferred route would be within 100 feet of 11 residential buildings and within 550 feet of 172 residential buildings. There are no commercial buildings within 100 feet of the route, but 14 commercials buildings within 550 feet. Two institutional buildings (hospitals, schools, churches, or government buildings) would be within 550 feet of the route, but none would be within 100 feet. Within 100 feet would be 17 outbuildings (sheds, garages, and barns) and within 550 feet would be 210 outbuildings.

Table 15: WE-GO's Preferred Route Buildings near ROW

Distance to ROW	Residential	Commercial	Institutional	Outbuildings
Within 100 feet	11	0	0	17
Within 550 feet	172	14	2	210

Transporting natural gas by pipeline involves some risk to the public in the event of an accident and release of gas. The company's preferred route was planned with consideration to public safety and reducing the inherent risk in the pipeline proximity. Considerations included limiting the number of buildings and residences within 220 yards of the pipeline, limiting the number of High Consequence Areas (such as nearby schools), limiting electric interference from electric transmission lines, limiting road ROW use (seven percent of Route A) to reduce the risk of excavation damage, and increasing access to pipeline ROW for leak surveys, maintenance, and emergency access.

In response to PSCW Data Request ELU-1.10[50] the applicant stated that Route A has eight high-consequence areas, Route B has six high-consequence areas, and the Preferred route has seven high-consequence areas. On Route A, there are 11.61 miles (35.2 percent) of Class 1

[50] PSC Ref#: 515240 Response-Data Request-PSC-Uphoff-1.10

locations, 19.59 miles (59.4 percent) of Class 2 locations, 1.11 miles (3.4 percent) of Class 3 locations, and zero miles in Class 4 locations. On Route B, there are 15.8 miles (47.8 percent) of Class 1 locations, 16.34 miles (49.5 percent) of Class 2 locations, 0.56 miles (1.7 percent) of Class 3 locations, and zero miles in Class 4 locations. On the preferred route, there are 15.71 miles (47.6 percent) of Class 1 locations, 16.06 miles (48.7 percent) of Class 2 locations, 1.2 miles (3.6 percent) of Class 3 locations, and zero miles in Class 4 locations. Class location is determined by population density along a pipeline segment, based on the number and proximity of residential or occupied structures. The potential impact circle where a failure could have significant impact on persons or property is 528 feet in radius.

High Consequence Areas, Moderate Consequence Areas, and Class locations, among other operational factors, determine the utility's long-term assessments of the pipeline under an integrity management program. Typically, the utility will run inline (internal) inspection tools every seven years to find damage, corrosion, coating or wall loss, and other anomalies. The utility must prioritize and address any anomalies under strict requirements. Each route plan includes construction of launching and receiving sites for inline inspection tools.

7. Evaluation of the Reasonable Alternatives and their Environmental Consequences

7.1 No-build alternative

Foregoing the construction of the proposed gas main and associated facilities would not meet WE-GO's stated need of providing additional firm deliverability of natural gas to southeastern Wisconsin, which is described in more detail in Section 2 of this EA. Presumably, this would also have implications for two other current proposals before the Commission in dockets 6630-CE-316, 6630-CE-317, and 6630-CG-140 and the associated gas supply needs. It is assumed that if this proposal were not constructed, another source of gas supply would still be needed, probably in a similar area. A more detailed discussion of the need for the project, along with data request information and analysis, can be reviewed in the additional documentation in the case documents for this docket on the Commission's Electronic Records Filing System (ERF) site on its website. If this project were not constructed, no environmental impacts would occur at this time.

7.2 Reasonable alternatives to the proposed project

WE-GO believes construction of the proposed transmission main is the best alternative and the most advantageous means of meeting their obligation as a public utility. Because this is a gas pipeline construction project, renewable resources and additional energy conservation and efficiency would not be cost-effective or technically feasible alternatives to the RLP project. The purpose of the project is to add firm deliverability of natural gas to southeastern Wisconsin; thus, currently available renewable technology cannot replicate the ability of a pipeline to deliver natural gas.

7.3 Alternative project routes

Several other project routes were considered before selecting a final proposed route. Factors considered while reviewing other possible routes included route location, construction costs, environmental impacts, and construction impacts. Table 8 of the company's application[51] shows how various routes were evaluated and moved because of landowner concerns. Section 2.7.2 of the application also explains why certain routes were reviewed but not chosen.

[51] PSC REF#: 518981 Rochester Lateral Project Supplemental CA Application

8. Wisconsin Environmental Protection Act Determination

When determining whether an EIS is warranted for a given Commission action, the Commission must consider ten broad factors listed in Wisconsin Admin. Code § PSC 4.20(2)(d). Based on the analysis provided in this EA, the following subsections provide Commission staff's conclusions regarding each of the ten factors with respect to the proposed project.

8.1 Effects on geographically important or scarce resources, such as historic or cultural resources, scenic or recreational resources, prime farmland, threatened or endangered species and ecologically important areas

The Commission must consider a proposed action's "[e]ffects on geographically important or scarce resources, such as historic or cultural resources, scenic or recreational resources, prime farmland, threatened or endangered species and ecologically important areas." Wis. Admin. Code § PSC 4.20(2)(d)1.

No geographically important or scarce resources would be significantly impacted within the area to be affected by construction and operation of the proposed project. Several threatened, endangered, or special concern species were identified to potentially occur in the project area. Impacts to these species could be mitigated or avoided altogether with the utilization of erosion control methods, directional drilling, additional species surveys, avoidance fencing, and/or time of year restrictions on construction work and vegetation removal, although there may be some impacts. Several segments intersect or adjoin public parks and recreational areas, as described in Section 5.11 of this EA. The applicant stated that BMPs would be used to minimize impacts to resources near or within any park or recreational area and existing ROW easements would also be used where possible to avoid any unnecessary clearing. Segment C-11 intersects Bong State Recreation Area and Brighton Dale Park. Segment B-9 could be used in this area to avoid intersecting park land.

Impacts to prime farmland are discussed in Section 5.8 of this EA. WE-GO confirmed that impacted agricultural land would be returned to original (pre-construction) condition and would be able to be farmed as normal after the pipeline is installed.[52] However, there are limited areas of agricultural land that would be permanently impacted because of the permanent easement, although farming operations could still continue over the pipeline. The applicant stated that Route A would contain 113.2 acres of permanent impacts, Route B would have 113.3 acres of permanent impacts, and Route C would contain 32.6 acres of permanent impacts. Overall, no significant impacts are expected to occur within scenic or recreational resources or ecologically important areas.

[52] PSC REF#: 528510 Response-Data Request-PSC-Thompson-EAT-2.05

8.2 Conflicts with federal, state, or local plans or policies

The Commission must consider a proposed action's "[c]onflicts with federal, state or local plans or policies." Wis. Admin. Code § PSC 4.20(2)(d)2.

There are no known conflicts of the proposed action with any federal, state, or local plans or policies. The proposed project's relation to Governor Evers' Clean Energy Plan[53] is described in section 5.13 above. While the proposed pipeline would contribute to the production of greenhouse gas emissions possibly beyond stated goals, it is also contemplated in energy transition plans that some level of greenhouse gas emitting sources would still be in existence for some period of time, while renewable energy resources and other technologies, programs, and efforts are still continuing to come online and be implemented.

8.3 Significant controversy associated with the proposed action

The Commission must consider any "[s]ignificant controversy associated with the proposed action." Wis. Admin. Code § PSC 4.20(2)(d)3.

The Commission received several public comments on the proposed project during the EA scoping period regarding concern about the gas pipeline in proximity to residential areas; the potential for a leak or explosion; the number of trees, natural areas, and species that would be disturbed; businesses that would be disrupted by construction; and impacts to climate change and human health. Comments such as these are typical during proceedings for natural gas pipeline projects. There is no known controversy regarding the type, magnitude, or significance of the environmental impacts expected to result from the proposed project.

8.4 Irreversible environmental effects

The Commission must consider "[i]rreversible environmental effects." Wis. Admin. Code §PSC 4.20(2)(d)4.

Construction of this natural gas main would not be expected to result in any truly irreversible environmental effects. Short-term impacts such as noise, air quality, erosion, and removal of vegetation would occur as a result of construction activities and would be irreversible. WE-GO confirmed that impacted agricultural land would be returned to original (pre-construction) condition and would be able to be farmed as normal after the pipeline is installed.[54]

Fossil fuels would be used during construction and the proposed project would contribute to emitting greenhouse gases during its operation. These environmental effects could not be reversed. As discussed above, the extraction and combustion of carbon-based fuels would add to the greenhouse gas buildup in the atmosphere and to the potential for global climate change. While discussions about potential global climate change impacts have been included to certain

[53] Wisconsin Office of Sustainability & Clean Energy (n.d.) Clean Energy Plan.
[54] PSC REF#: 528510 Response-Data Request-PSC-Thompson-EAT-2.05

extents in some Commission environmental documents, global climate change impacts encompass very complex considerations that are greater in scope than this project review. It is not clear if the effects of global climate change may at some point become reversible.

8.5 New environmental effects

The Commission must consider "[n]ew environmental effects." Wis. Admin. Code § PSC 4.20(2)(d)5.

There are no unique or new environmental effects expected as a result of this project. The proposed project utilizes common pipeline construction techniques (trenching and HDD) that are used when constructing natural gas pipelines. However, any emissions that would be produced by the new facilities connected to this pipeline would be new environmental effects because emissions from those facilities did not previously exist. There could possibly be some new, noticeable agricultural field impacts that could linger after construction where the pipeline was placed due to any subsidence, wet spots, or lingering compaction that could affect crop production.

8.6 Unavoidable environmental effects

The Commission must consider "[u]navoidable environmental effects." Wis. Admin. Code § PSC 4.20(2)(d)6.

As discussed in this EA, construction of the pipeline would result in a range of environmental effects that could not be avoided by route selection or construction methods. Some effects may be reduced or minimized but would not be entirely eliminated as a result of project activities.

Examples of unavoidable effects that would occur during construction include, but are not limited to:

- Impacts to local traffic patterns and roads, particularly if heavy/oversized vehicles are delivering materials;
- Soil and vegetation disturbance during trenching, drilling, and installation of the pipeline;
- Noise and vibration impacts due to project machinery and construction activities;
- Air quality impacts as a result of diesel fumes and dust during construction;
- Soil erosion, storm water runoff during precipitation events, particularly if there are loose soils from construction; and
- Disturbance to wildlife.

Certain unavoidable impacts caused by the proposed project that would be longer-term due to maintenance of the pipeline. These are anticipated to include:

- Impacts to air quality from greenhouse gases and other pollutants, regulated by state and federal permitting; and
- Permanent removal of trees and woody vegetation within a 20-foot easement directly over the pipeline.
- Permanent impacts as identified in PSC REF#: 538345[55]

8.7 Precedent-setting nature of the proposed action

The Commission must consider "[t]he precedent-setting nature of the proposed action." Wis. Admin. Code § PSC 4.20(2)(d)7.

The proposed project would not set any precedents. The proposed project is a common activity of natural gas utilities. The Commission has reviewed a range of natural gas pipeline projects in the past, and this type of proposed project falls within a similar scope to utility infrastructure projects previously reviewed and authorized.

8.8 Cumulative effect of the proposed action when combined with other actions and the cumulative effect of repeated actions of the type proposed

The Commission must consider "[t]he cumulative effect of the proposed action when combined with other actions and the cumulative effect of repeated actions of the type proposed." Wis. Admin. Code § PSC 4.20(2)(d)8.

This proposed pipeline project is one of multiple related We Energies construction applications that are currently under review by the Commission at the time of writing this EA. The RLP would provide additional required firm natural gas service to Wisconsin Electric's proposed Oak Creek Combustion Turbine (OCCT) generation facility (Docket ID 6630-CE-317),[56] Paris Reciprocating Internal Combustion Engine (RICE) generation facility (Docket ID 6630-CE-316)[57], and subsequently to the Elm Road Generating Station (ERGS) after enhancements are made that would allow ERGS to operate completely fueled by natural gas. Two pipelines are also planned to be constructed from the Rochester Lateral to the proposed Oak Creek Liquified Natural Gas (OCLNG) project (Docket ID 6630-CG-140)[58] as part of this project. As stated in the application, "RLP is a critical piece of infrastructure that is part of a multi-pronged approach, along with firm interstate pipeline capacity (ANR 2027 Expansion) and the OCLNG facility."

The environmental impacts and need analyses associated with the Paris RICE, OCCT, and OCLNG projects are described in the respective EIS and EAs and other materials submitted to their respective dockets. Greenhouse gas emissions produced by the proposed generation facilities would be a contributor to the overall effect of global climate change. There has been a

[55] PSC REF#: 538345 Response-Data Request-PSC-Ingwll-VBL-03242025 Attach 01
[56] PSC Docket 6630-CE-317
[57] PSC Docket 6630-CE-316
[58] PSC Docket 6630-CG-140

recent trend in the United States to lower overall carbon emission as part of an overall energy transition. The energy transition at this time largely involves increasing the use of renewable energy sources along with the continued use and construction of some traditional carbon-emitting facilities such as the proposed project, to still allow for a reliable electric system.

8.9 Foreclosure of future options

The Commission must consider "[t]he foreclosure of future options." Wis. Admin. Code § PSC 4.20(2)(d)9.

The proposed pipeline would not foreclose any future environmental or land uses in the project area; however, easements typically have restrictions that prevent certain activities, excavations, or the building of structures within the easement. Typically, no structures, water retention, or septic systems are permitted within the immediate easement area, and there would likely be restrictions on the type or quantity of woody vegetation allowed to grow directly over the pipeline. Agricultural operations should be able to resume after installation.

8.10 Direct and indirect environmental effects

The Commission must consider "[d]irect and indirect environmental effects." Wis. Admin. Code § PSC 4.20(2)(d)10.

Direct environmental effects of the proposed project would include short-term, localized effects from construction activities. Long-term direct environmental effects would include tree clearing and maintenance of existing and/or new ROW areas. As discussed in Section 5.8 of this EA, research has shown that there may be long-term effects related to soil disturbance and crop yield loss on agricultural land.

Construction activities would disturb the vegetation and soils along the route of the gas main project, primarily in areas where trenching is used for placing the pipe. Tree clearing and similar environmental effects of the proposed project could possibly lead to changes in habitat and species compositions along the project route areas. During excavation activities, there would be an increased risk of soil erosion, which in areas near wetlands or waterways is a serious concern with regard to sedimentation. Erosion control methods can decrease this risk. Properly undertaken site restoration after pipeline installation would allow the soils and vegetation disturbance to be temporary in nature. During the growing season, any areas of disturbed soils create a high risk of invasion by non-native invasive plants. Loose soils should be stabilized with non-invasive cover crops as soon as possible, and machinery or equipment cleaned in accordance with terrestrial or aquatic invasive species BMPs as applicable.

Community impacts would be primarily related to impacts to roads and driveways in the project area. Residents using local roadways during construction of the project would experience lane closures and to a lesser extent minor traffic congestion. These impacts should be short-term and managed through municipal permitting of lane closures or diversions. During construction activities, there would be increased noise, dust, and vibration in the construction areas. These effects would cease once construction is complete.

Indirect effects of the project would include the environmental impacts and air emissions from extracting and transporting natural gas as the fuel source for the proposed pipeline, the construction and transportation system inputs for making and transporting all of the materials and ongoing fuel needed for this proposed project. There are a range of indirect environmental impacts such as the mining of frac sand and impacts to water and air from the process of fracking for natural gas, if that is how the natural gas would be extracted. The proposed pipeline would indirectly contribute to greenhouse gas emissions, which is the driving factor of climate change. The negative effects of greenhouse gas emissions and climate change on human health and welfare are well documented, as discussed in Section 5.13 of this EA.

Any change in plant composition or habitat could have an effect on wildlife or other species usage or occupancy of the areas along the project. Certain types of land usage over the pipe would no longer be available due to the restrictions placed on land usage in a pipeline easement.

9. Conclusion

This EA informs the Commissioners, the affected public, and other interested people about the proposed Rochester Lateral Project and its potential environmental and societal impacts. Through data requests, additional analyses, and a review of public comments, Commission staff has provided thorough, factual, and up-to-date information about the project, potential impacts of the proposed project, and mitigation measures that could address some potential impacts. The EA concludes that construction and operation of the project would be likely to have a range of environmental effects. However, Commission staff has not identified any potential environmental effects of the proposed project that could be considered significant. This evaluation is arrived at assuming that some, if not all, of the mitigation measures proposed by the applicant, Commission, and DNR are used.

Where no disagreements or controversies otherwise exist in relation to the proposed construction plan and BMPs, prior Commission orders for projects include a condition that the applicant construct the project according to the application and all data requests and responses, including any environmental mitigation strategies. There are no current disagreements or controversies relating to the mitigation measures proposed by the applicant, and it is therefore reasonable to assume they would enact these as described if the project is approved.

The potential impacts of the project are described in this EA so that members of the public can use this information to better inform their comments upon the Preliminary Determination; and so that the Commission can address the potential impacts and mitigation possibilities in its deliberations about project certification. Because potential impacts and major concerns can be substantially addressed through construction and operational mitigation measures committed to by the applicant or are expected to be regulated according to standards designed to protect public health and other natural resources, no significant impacts on the human environment are expected to occur as a result of this project.

This assessment finds that possible approval and construction of this project would be unlikely to have a significant impact on the human environment as defined by Wis. Stat. § 1.11, therefore the preparation of an EIS is not required.

10. Recommendation

__X__ No significant impact. Environmental review complete. Preparation of environmental impact statement is not necessary.

_____ Prepare an environmental impact statement.

Submitted by: __Emily A. Thompson_____
Title: __Environmental Analysis and Review Specialist – Adv.
Date: __April 25, 2025_____

This environmental assessment complies with Wis. Stat. § 1.11, and Wis. Admin. Code § PSC 4.20.



Adam Ingwell
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Division of Digital Access, Consumer and Environmental Affairs

Date: May 20, 2025

Appendix A – List of Public Comments Received During EA Notification Period

Document	Received Date	Description
521119	10/15/2024	Public Comment by Scott Mason
521418	10/17/2024	Pete Beth Public Comment
525375	12/1/2024	Public Comment by Jacob Tombari
525990	12/6/2024	Public Comment by Jack Abell
526327	12/11/2024	Public Comment by Lawrence J. Schneider
526958	12/17/2024	Public Comment by Chris Van Eimeren
527456	12/19/2024	Public Comment by Gail Kudla
527853	12/20/2024	Public Comment by Thomas R Klopf
527961	12/22/2024	Public Comment by Steven Kopczynski
528000	12/23/2024	Public Comment by Jenny Hoople
528033	12/23/2024	Public Comment by Healthy Climate Wisconsin
528039	12/23/2024	Public Comment by Scott Leibbrandt
528064	12/23/2024	Public Comment by Brittany Keyes
528192	12/27/2024	Public Comment - Richard Scott
531592	2/3/2025	Public Comment - William Meyers & Caitlin Forsythe

Appendix B – List of Public Comments Received During Preliminary Determination Period

Document	Received Date	Description
544336	5/1/2025	Public Comment by Vitus Hloushek
545669	5/12/2025	Public Comment by Siana L Loos
545670	5/12/2025	Siana Loos PD letter comment
545818	5/13/2025	Public Comment by Edward M. Usher
545857	5/14/2025	Public Comment by Rich Sudlow
545900	5/14/2025	Public Comment by J Waldron
545901	5/14/2025	Public Comment by Lynn Schmidt
545902	5/14/2025	Public Comment by Judith Stadler
546070	5/16/2025	Public Comment by Gianna Fernandez
546098	5/16/2025	Sarah Kopesky PD letter comment
546120	5/16/2025	Public Comment by Reetu Gupta
546124	5/16/2025	Public Comment by Catherine Frerker
546125	5/16/2025	Public Comment by Jessica Hannebert
546139	5/16/2025	Comments from 7 Health, Justice, and Environmental Groups
546150	5/16/2025	Public Comment by Meghan Pierce
546206	5/16/2025	Walnut Way comments on draft environmental assessment
546210	5/16/2025	Public Comment by Henry Mongrain
546211	5/16/2025	Public Comment by Paul Kempf
546212	5/16/2025	Public Comment by Paul Kempf